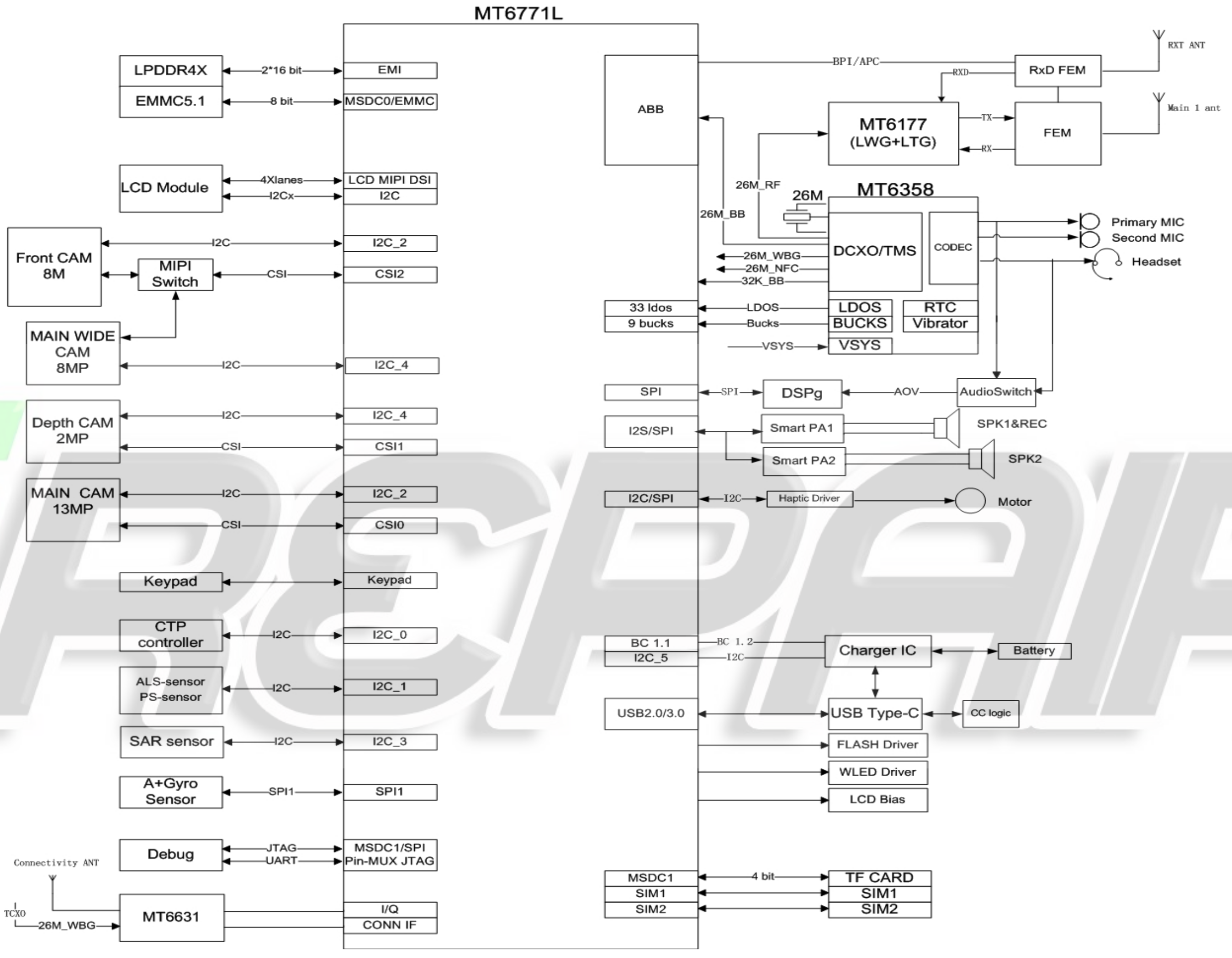
Riga L System Function Block Diagram



Title	Page Name = 00_Block Diagram	REV: V10	
DOCUMENT NO.:	Design Name = 98870_1_11_201906141138	Size D	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = YANGSHAOYU
Date:	Page Modify Date = Friday, June 14, 2019	Sheet 1 of 65	

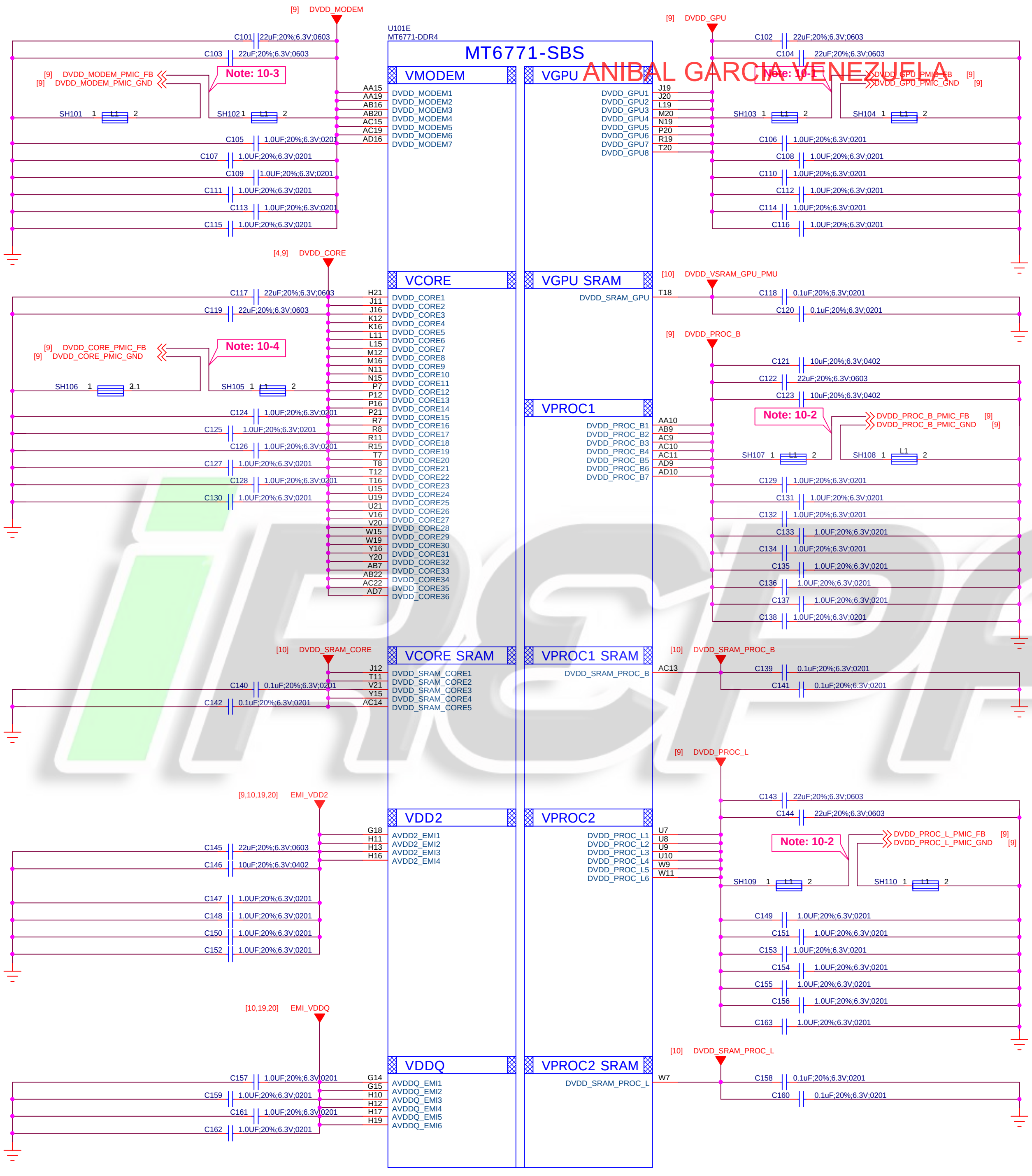
I2C	AP, SCP, SSPM	Function	I2C Spec.	Budget Timing	I2C Address
I2C-0	AP	CTP	400 Kbps	Yes.	ILI9881H I2C address: Write:0xAA, Read:0xAB
I2C-1(I3C)	AP ,SCP	ALS and P Sensor	400 Kbps	Yes.	TMD27504 I2C Address : Write: 0x92, Read: 0x93 MN58406DKDN I2C Address : Write: 0x82, Read: 0x83
		Ambient Light Sensor	400 Kbps	Yes.	TSL2560 I2C address: (Write:0x72, Read:0x73 MN26005UKDN I2C address: Write:0x92, Read:0x93
I2C-2 (I3C)	AP	Front Camera - 8M	400 Kbps	Yes.	S5K4H7YX03-FGX9 I2C address: Write:0x20, Read:0x21
		Main Camera - 13M	400 Kbps	Yes.	S5K3L6XX03-FGX9 I2C address: Write:0x5A, Read:0x5B
I2C-3	AP	HAPTIC DRIVER	400K bps	Yes.	DRV2624 I2C address: Write:0xB4, Read:0xB5 AW8624 I2C address: Write:0xB4, Read:0xB5
		Sar Sensor	400K bps	Yes.	A96T346DFP I2C address: Write:0x40, Read:0x41
		Smart PA1	400K bps	Yes.	TAS2563 I2C address: Write:0x9A, Read:0x9B
		Smart PA2	400K bps	Yes.	TAS2563 I2C address: Write:0x9C, Read:0x9D
I2C-4 (I3C)	AP	Wide Camera - 8M	400 Kbps	Yes.	S5K4H7YX03-FGX9 I2C address: Write:0x20, Read:0x21
		Depth Camera - 2M	400 Kbps	Yes.	OV02A10-GA4A I2C address: Write:0x7A, Read:0x7B
I2C-5	SSPM	Charger	400 Kbps	Yes.	RT9471D I2C address: Write:0xA6, Read:0xA7
		CC Logic	400 Kbps	Yes.	TUSB320 I2C address: Write:0x8E, Read:0x8F RT1711 address: Write:0x9C, Read:0x9D
		LCD Bias	400 Kbps	Yes.	SM5109 I2C address: Write:0x7C, Read:0x7D OCP2131 I2C address: Write:0x7C, Read:0x7D

Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

SPI

SPI List

	GPIO	Device	Device IC	IC供应商
SPI0	GPIO[85;86;87;88]	Fingerprint	FPC1510A	
			/	/
SPI1	GPIO[161;162;163;164]	A+G	ICM-40607	INVENSENSE
			BMI160	BOSCH
SPI2	GPIO[0;1;2;94]	LCM	ILI9881H	ILITEC
SPI3	GPIO[21;22;23;24]	CODEC(DSP)	DBMD4	DSPG
SPI4	GPIO[4;5;6;7]			



Schematic design notice of "10_BB_POWER_PDN" page.

Note 10-1: Differential pair of DVDD_GPU remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-2: Differential pair of DVDD_PROC remote sense must be close to BB's ball
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-3: Differential pair of DVDD_MODEM remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Note 10-4: Differential pair of DVDD_CORE remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

Title	01_BB_POWER_PDN		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
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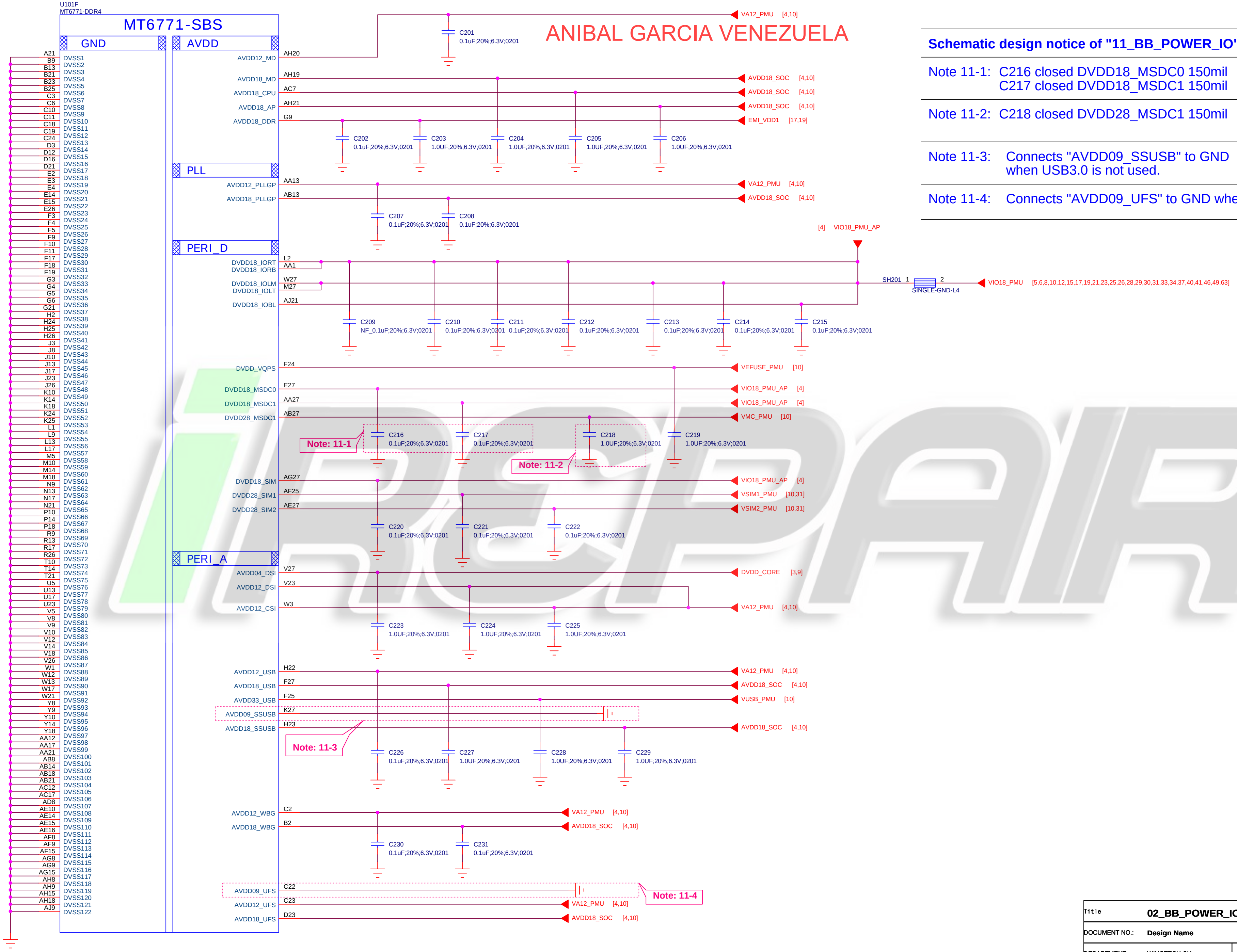


Schematic design notice of "11_BB_POWER_IO" page.

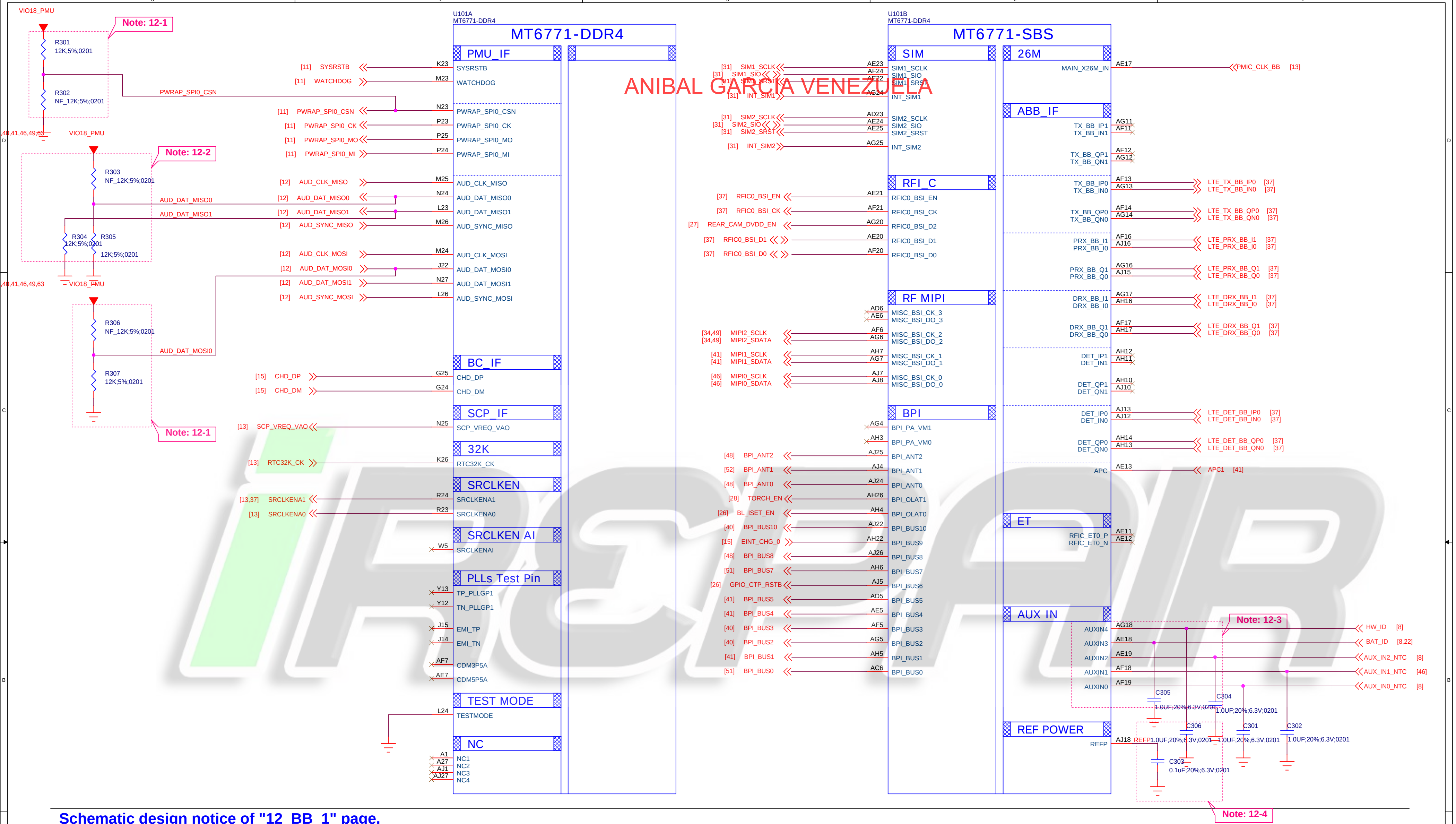
Note 11-1: C216 closed DVDD18_MSDC0 150mil
C217 closed DVDD18_MSDC1 150mil

Note 11-2: C218 closed DVDD28_MSDC1 150mil

Note 11-3: Connects "AVDD09_SSUSB" to GND when USB3.0 is not used.

Note 11-4: Connects "AVDD09_UFS" to GND when UFS is not used.

Title	02_BB_POWER_IO	REV: V10	
DOCUMENT NO.:	Design Name	Size	C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Friday, June 14, 2019	Sheet 4 of 65	



Schematic design notice of "12_BB_1" page.

Note 12-1: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pins to select which interface will be the JTAG pin out.

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	AP_JTAG	IO_JTAG
HI	LO	N/A	N/A
HI	HI	SPI_CSB/SPI_CLK/ SPI_MO/SPI_MI/EINT8	N/A
LO	LO	SPI_CSB/SPI_CLK/ SPI_MO/SPI_MI/EINT8	DPI_11/DPI_HSYNC/DPI_VSYNC/DPI_DE/ DPI_CK/DPI_D8/DPI_D9
LO	HI	MSDC1_CLK/CMD/ DAT0/DAT1/DAT2	N/A

Note 12-2: "AUD_DAT_MISO0" is bootstrap pin to enable serial JTAG output over USB2.0 interface or not.
When "AUD_DAT_MISO0" is pulled to high in system start up and then USB2.0 interface will be switched into serial JTAG mode.

"AUD_DAT_MISO1" is bootstrap pin to select system booting up from eMMC or UFS device.

AUD_DAT_MISO1	Booting device
LO	eMMC
HI	UFS

Note 12-3: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-4: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.

Note 12-5: AUD_SYNC_MISO and AUD_CLK_MISO are DDR type feature in bootstrap

AUD_SYNC_MISO	AUD_CLK_MISO	DDR
LO	LO	LPDDR4X
LO	HI	LPDDR4X(Ext x 2 EN)
HI	LO	LPDDR3
HI	HI	LPDDR4X(Ext x 1 EN)

Title	03_BB_1_RF&SIM_IF	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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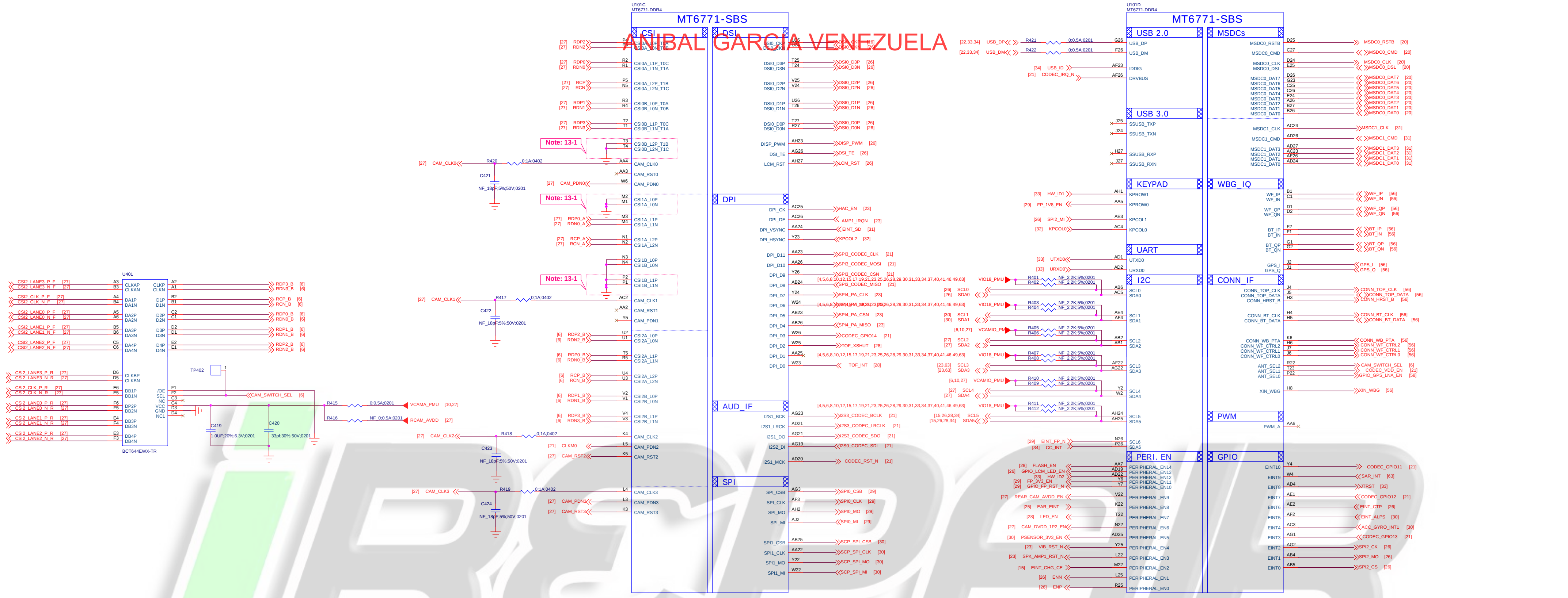

Schematic design notice of "13_BB_2" page.

Note 13-1: CSI ports which are no use could be connected to GND or set in NC.
For detail information, please refer to MT6771 Design Notice

Title	04_BB_2_MIPI&GPIO	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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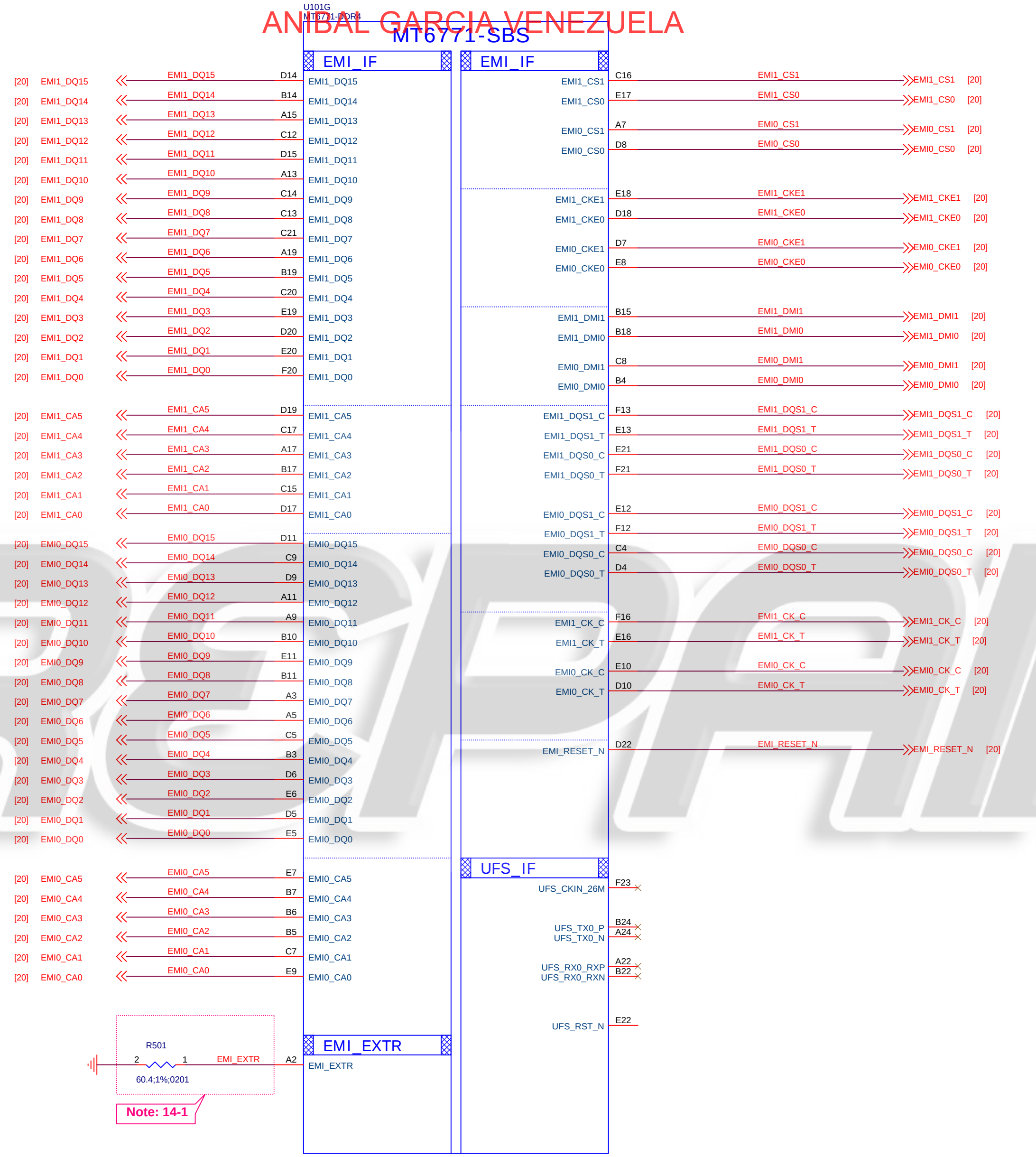

Schematic design notice of "14_BB_3" page.

Note 14-1: The resistor of EMI_EXTR for DRAM has to be placed near to BB as close as possible R501, please select 60.4 ohm 1% resistor

Title	10_BB_ POWER_PDN	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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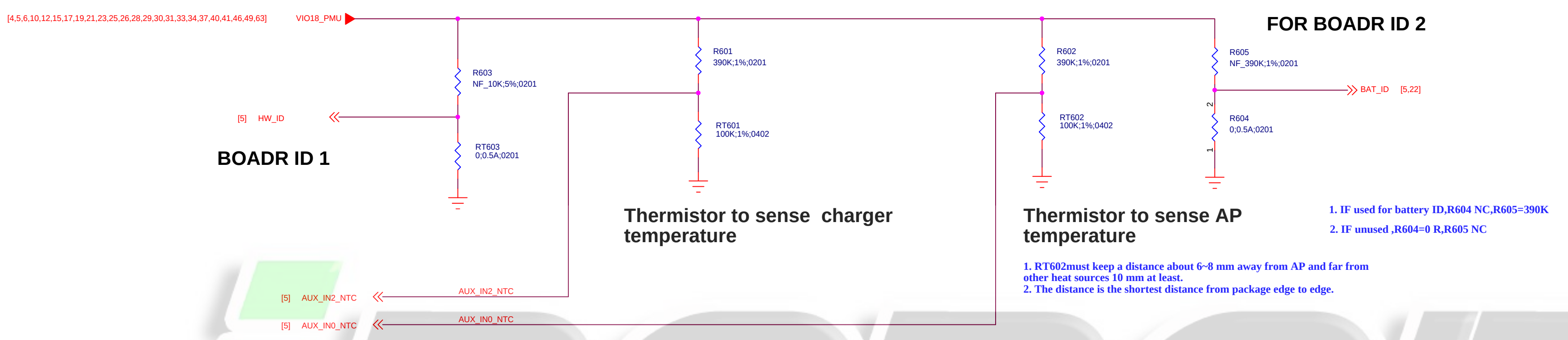

Title	06_BB_AUXADC_Thermal	REV: V10
DOCUMENT NO.:	Design Name	Size C
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH		
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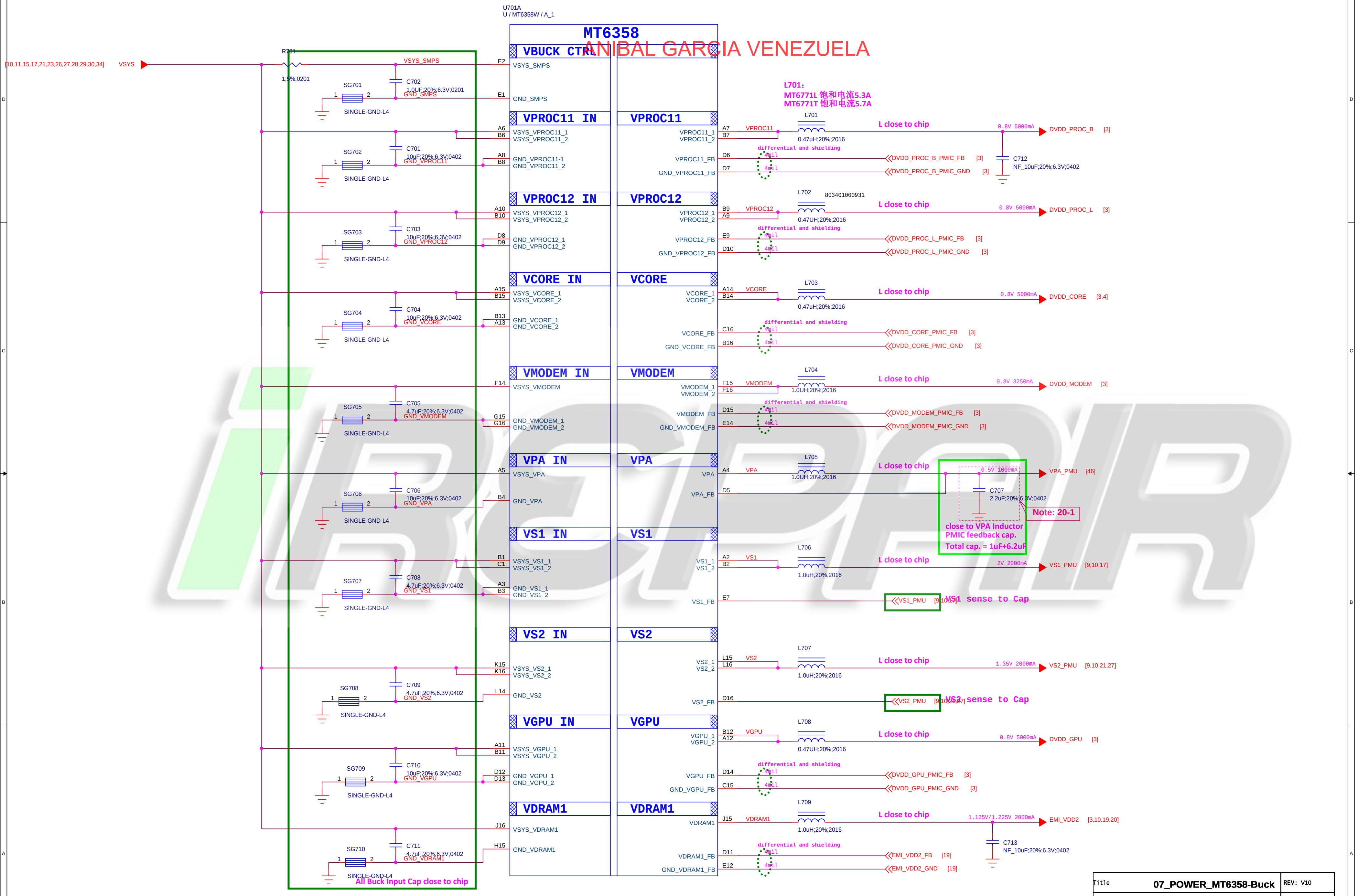

Title	07_POWER_MT6358-Buck	REV: V10
DOCUMENT NO.:	Design Name	Size C
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH		
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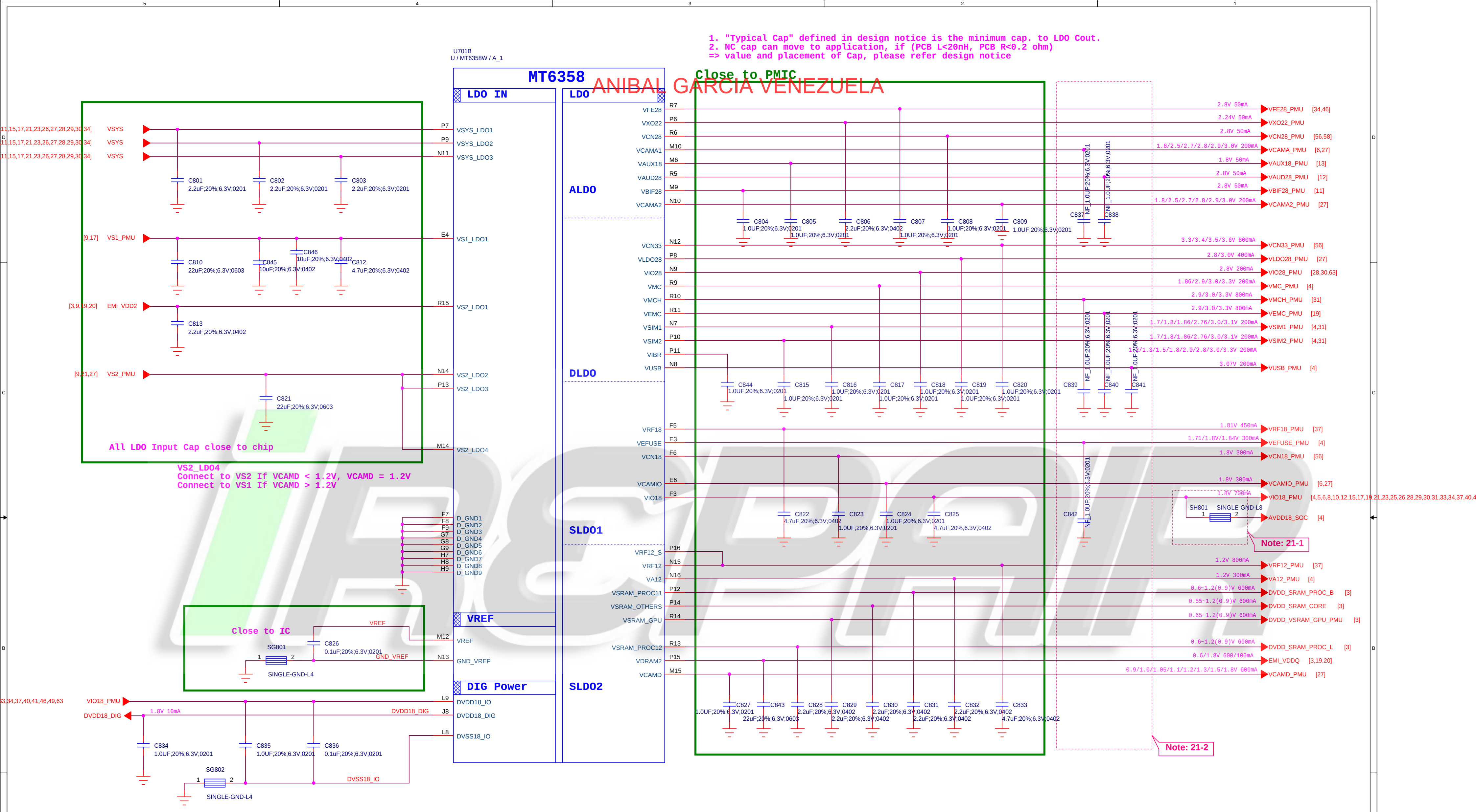
Schematic design notice of "20_POWER_MT6358-Buck" page.

Note 20-1: Please select C707 with 0402 size



Schematic design notice of "21_POWER_MT6358-LDO" page.

Note 21-1: Please set SH801 close to C825, making star connection between VIO18_PMU and AVDD18_SOC near to LDO cap. C825
Please also refer to MT6358 design notice for further detail design information

Note 21-2: If these power trace can meet LDO layout constraint, these CAP can be NC or removed.
Please refer to MT6358 design notice.

Title	08_POWER_MT6358-LDO	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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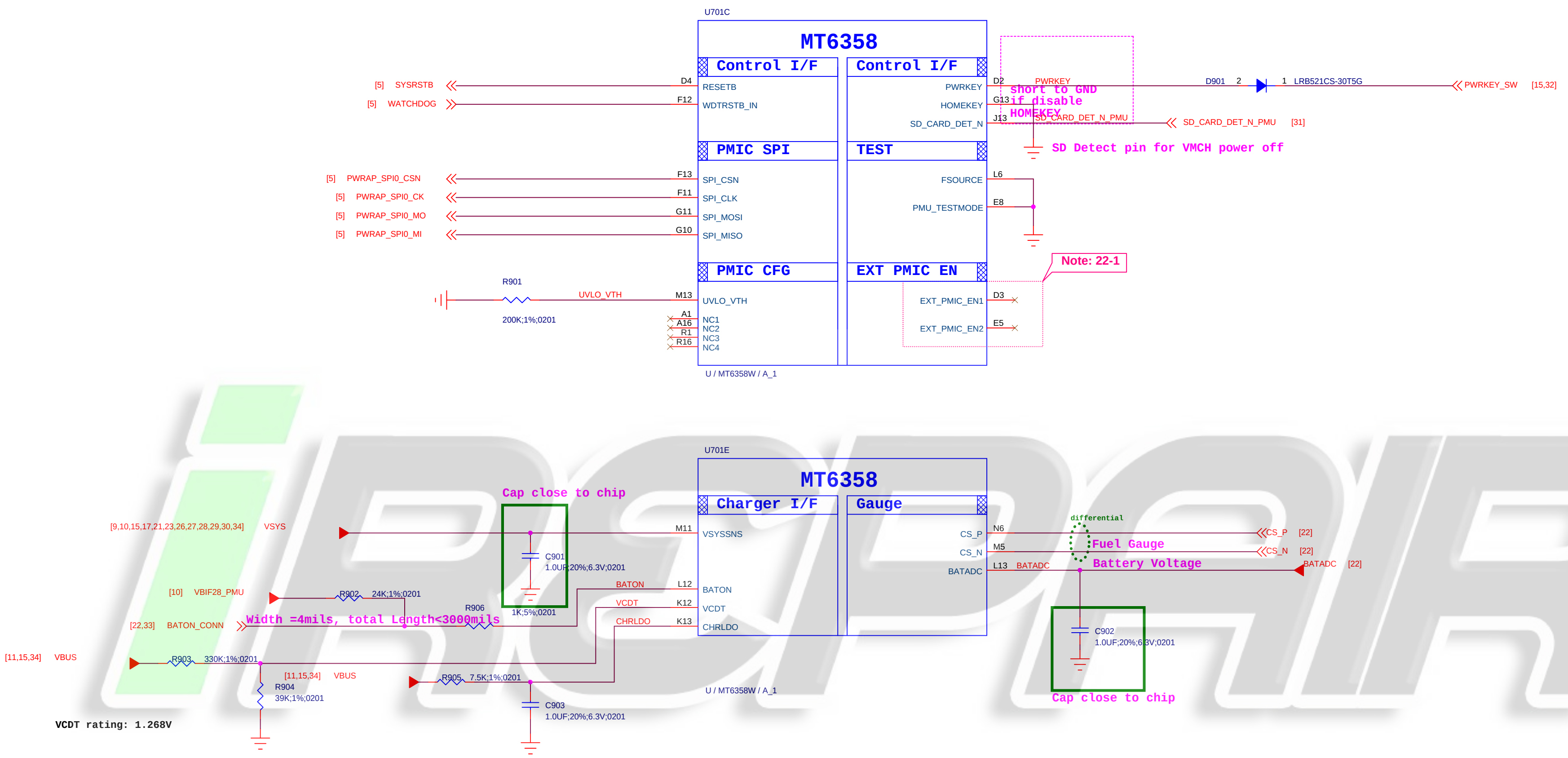

Schematic design notice of "22_POWER_MT6358-General"

Note 22-1: EXT_PMIC_EN1 : For UFS_1V8, and keep floating if it is not used
EXT_PMIC_EN2 : For VA09 of SSUSB/UFS, and keep floating if it is not used

Title	09_POWER_MT6358-General		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
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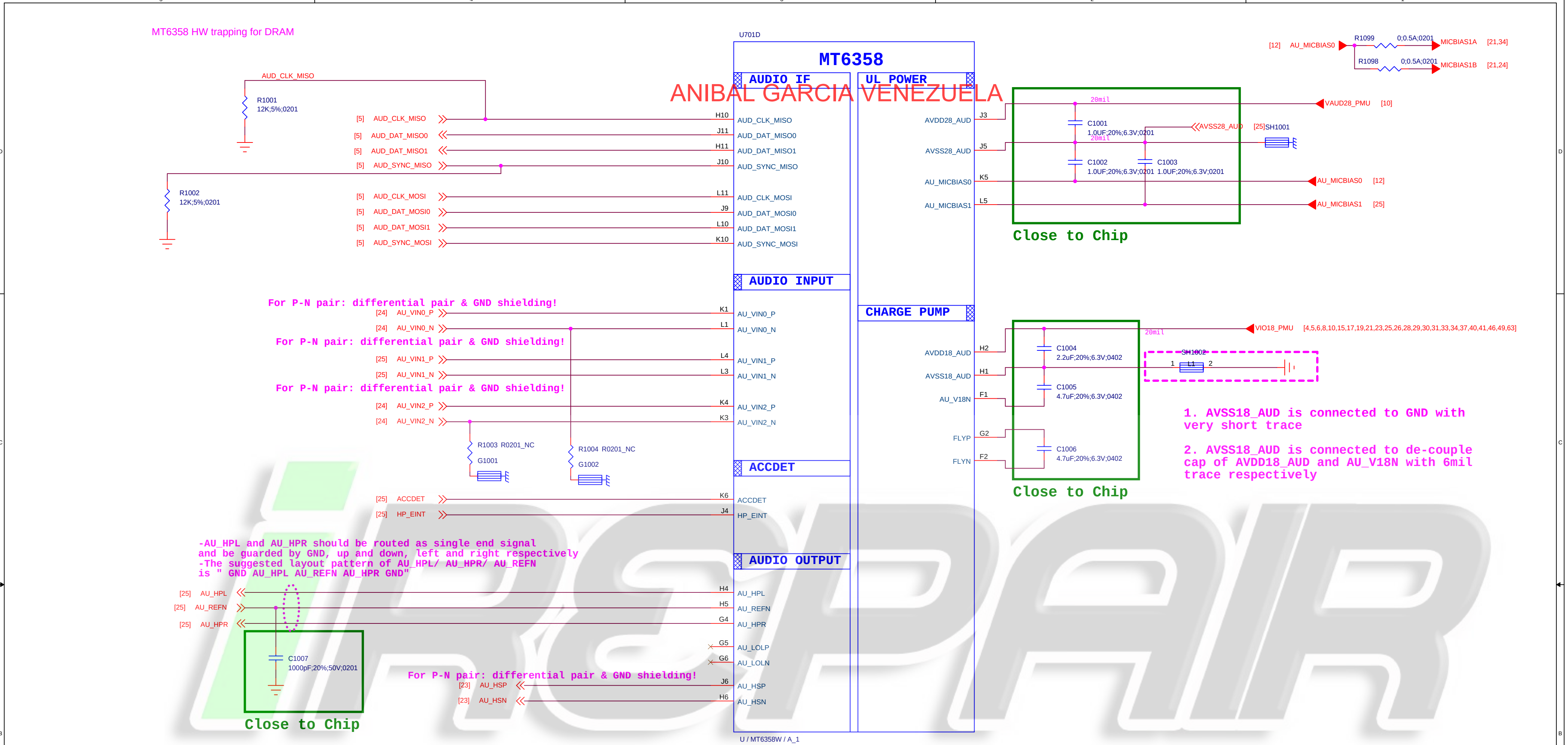

Schematic design notice of "23_POWER_MT6358-Audio"

Note 23-1: VDRAM 2 / VDRAM1 output voltage vs. trap pin.

HW GPIO configuration		Trapping Option		DRAM type	VDRAM2 Power source (VS2_LDO1_ball)
AUD_SYNC_MISO	AUD_CLK_MISO	VDRAM1	VDRAM2		
0	0	1.125V	0.6V	LP4X	VDRAM1
0	1	OFF	1.8V	LP4X (Ext x 2 EN)	VS1
1	0	1.225V	OFF	LP3	VDRAM1
1	1	1.125V	1.8V	LP4X (Ext x 1 EN)	VS1

Title	10_POWER_MT6358-Audio	REV: V10	
DOCUMENT NO.:	Design Name	Size C	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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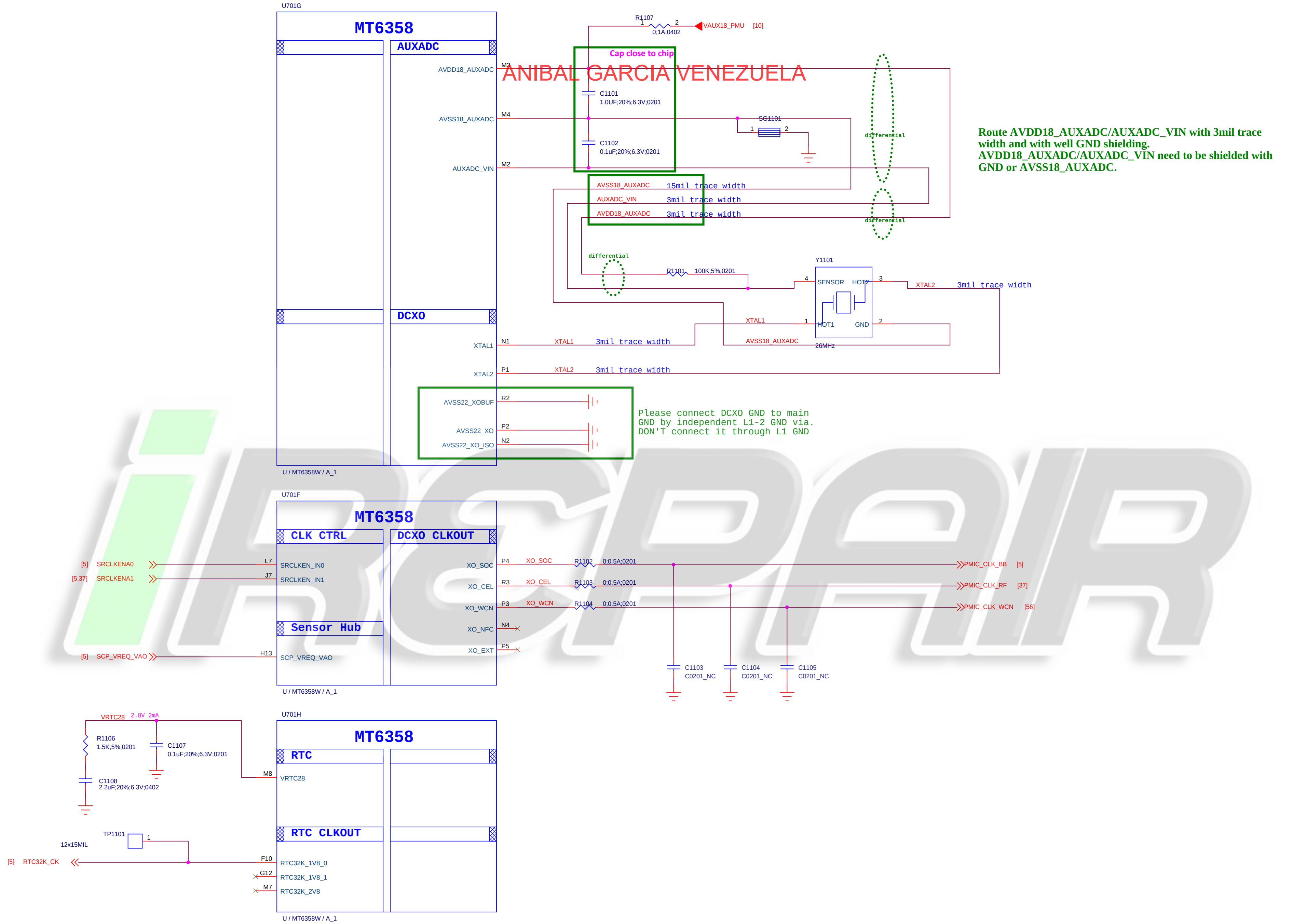

Title	11_POWER_MT6358_Clock		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
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Title	12_POWER_MT6370-General		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
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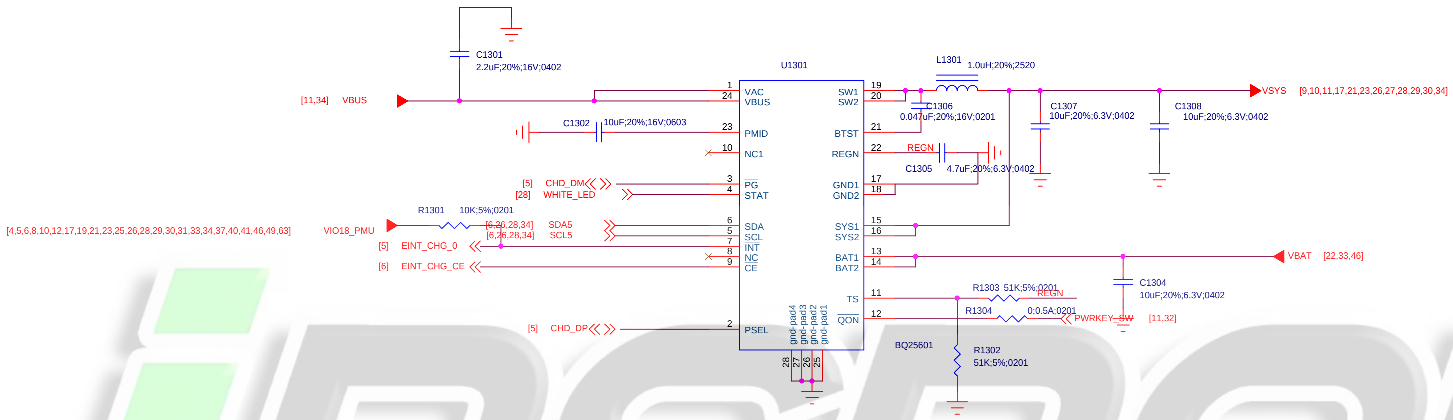

Title	13_POWER_MT6370-Charger + PP	REV:	V10
DOCUMENT NO.:	Design Name	Size	C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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Schematic design notice of "27_POWER_SubPMIC-HV powers" page.

Note 27-1: It is recommended to reserve 0-ohm and cap. for BOM fine tune to minimize RF de-sense.

Note 27-2: It is recommended to reserve 0-ohm for BOM fine tune to minimize RF de-sense.

Title	14_POWER_MT6370-HV powers		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
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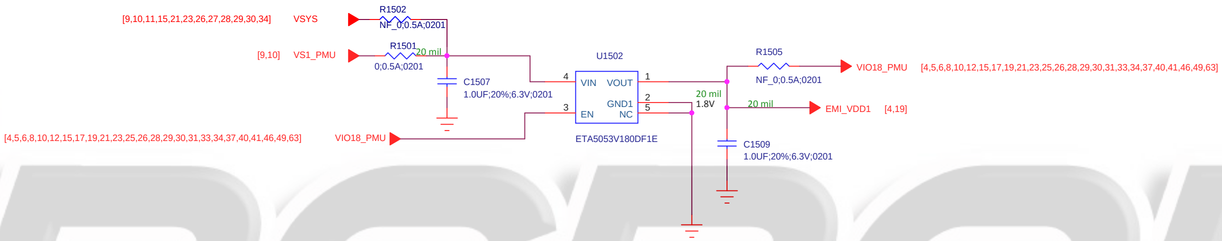
LDO for EMI_VDD1 of LPDDR4 VDD1



Title	15_POWER_ThirdParty_Powers	REV: V10
DOCUMENT NO.:	Design Name	Size C
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH		
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Title	16_BB_ POWER_PDN		REV: V10
DOCUMENT NO.:	Design Name		Size C
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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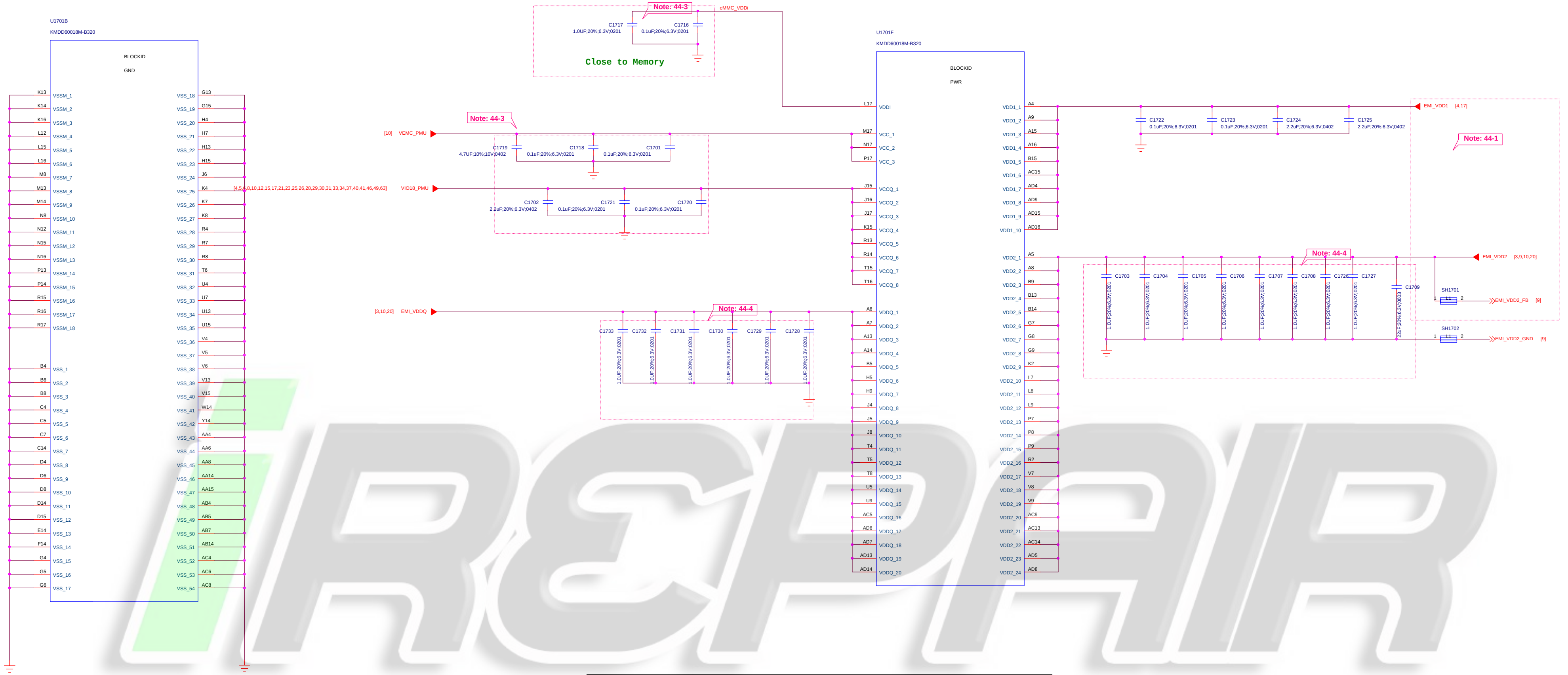

Schematic design notice of "44_Memory_eMMC_LPDDR4X"

Note 44-1: Please refer to power supply related page select VDRAM 2 / VDRAM1 output voltage properly for LPDDR4X

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,

Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.

Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC [>10uF, at PMIC page]:
please also refer to MMD and layout guide for placement.

Title	17_Memory_LPDDR4_POWER	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI	
WINGTECH			
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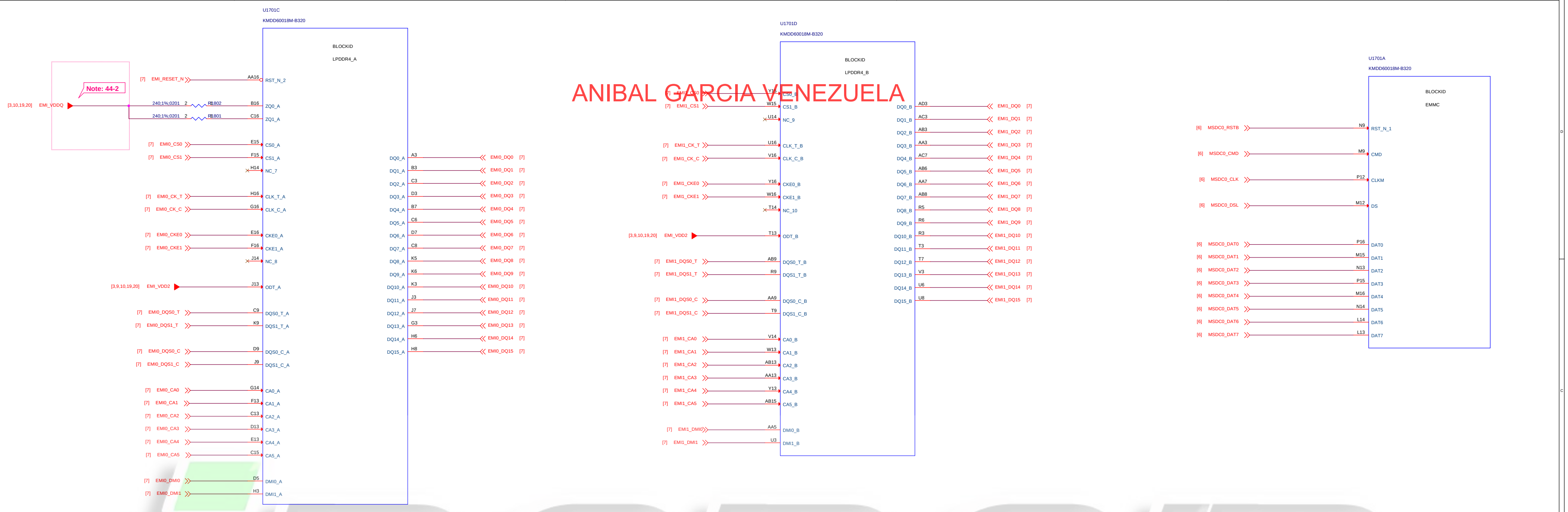

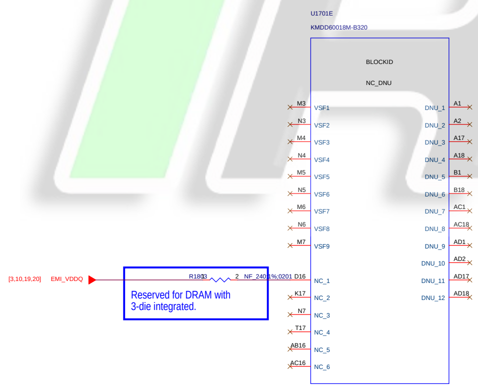

Schematic design notice of "44_Memory_eMMC_LPDDR4X"

Note 44-1: Please refer to power supply related page select VDRAM 2 / VDRAM1 output voltage properly for LPDDR4X

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ,

Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.

Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball. For other cap for PMIC [>10uF, at PMIC page]: please also refer to MMD and layout guide for placement.

Title	18_Memory_eMMC_LPDDR4		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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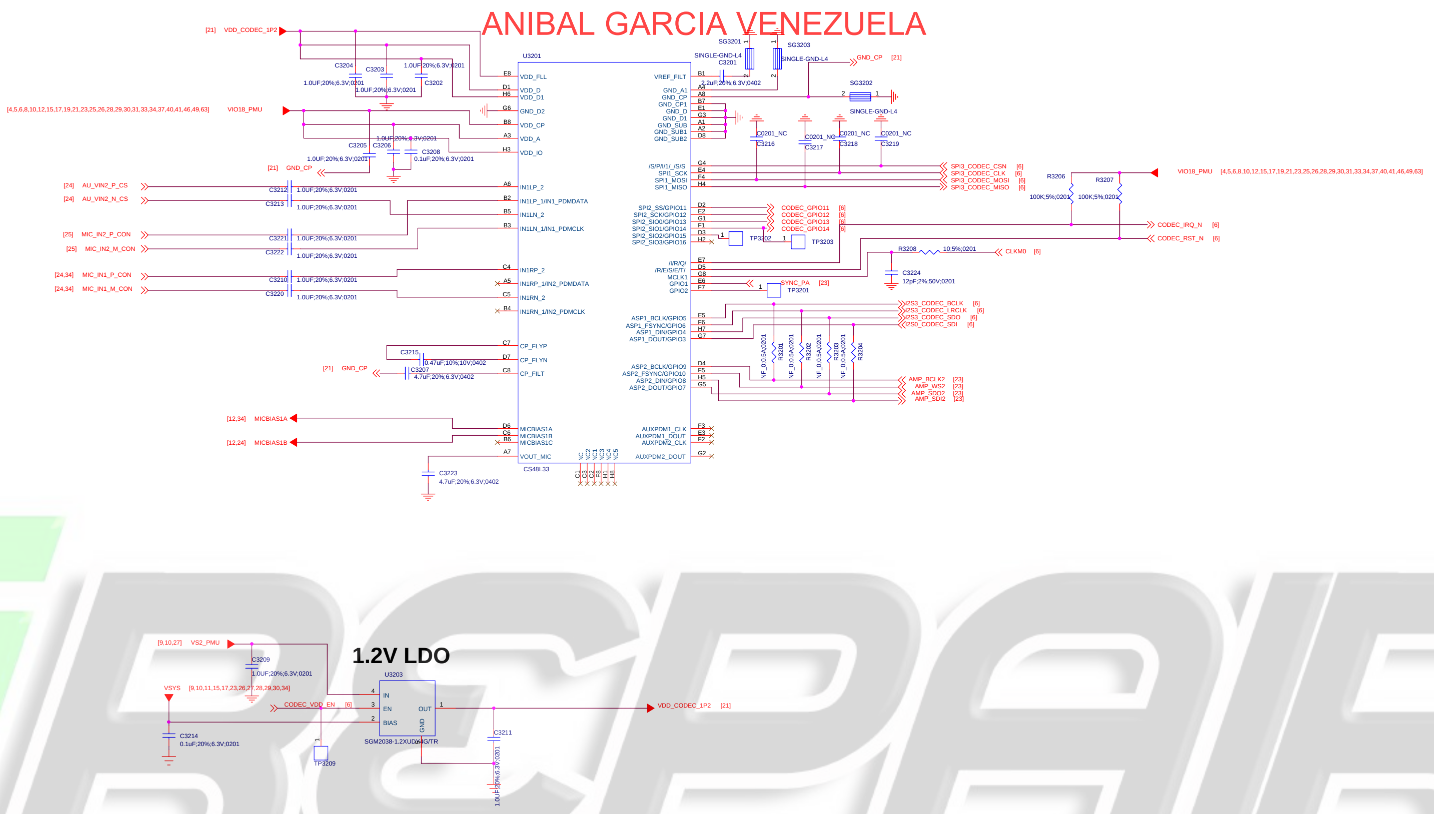

Title	32.CODEC_Cirrus	REV:	V10
DOCUMENT NO.:	98870_1_11_201906141113	Size	D
DEPARTMENT:	DEPARTMENT	DESIGNER:	DESIGNER
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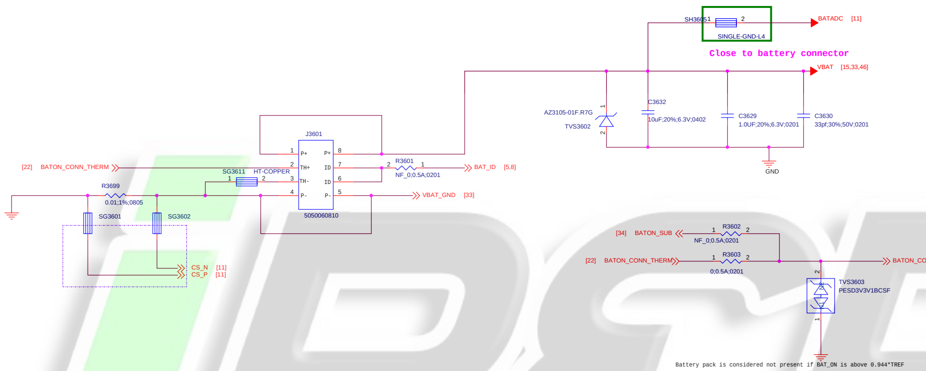
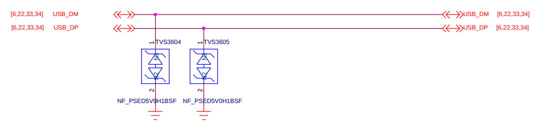
BATTERY CONNECTOR





Title	36_Battery/USB IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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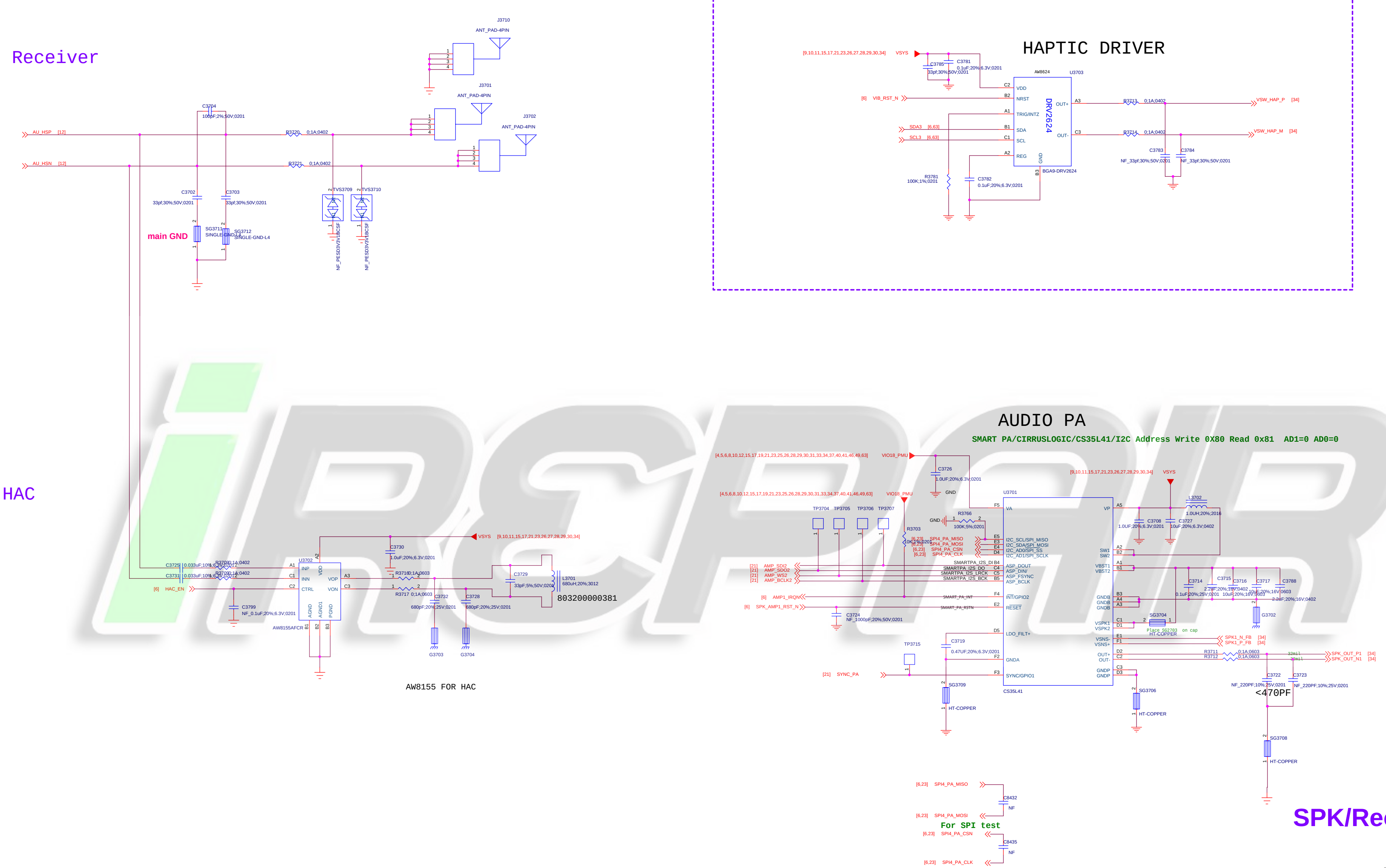

SPK/Receiver/Vibrator

Title	37_SPK/Receiver/Vibrator	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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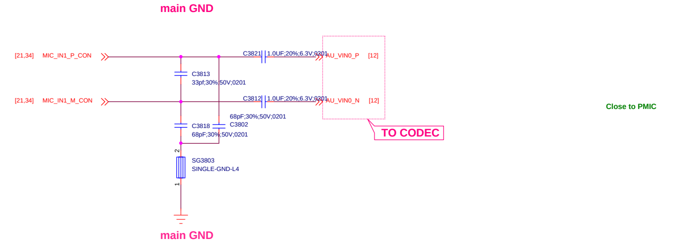
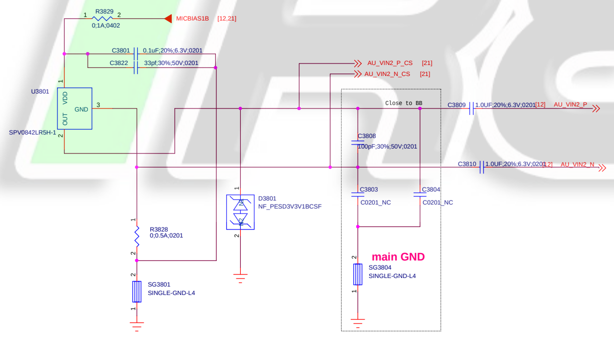
MAIN MIC



TOP MIC



Title	38_MIC	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI
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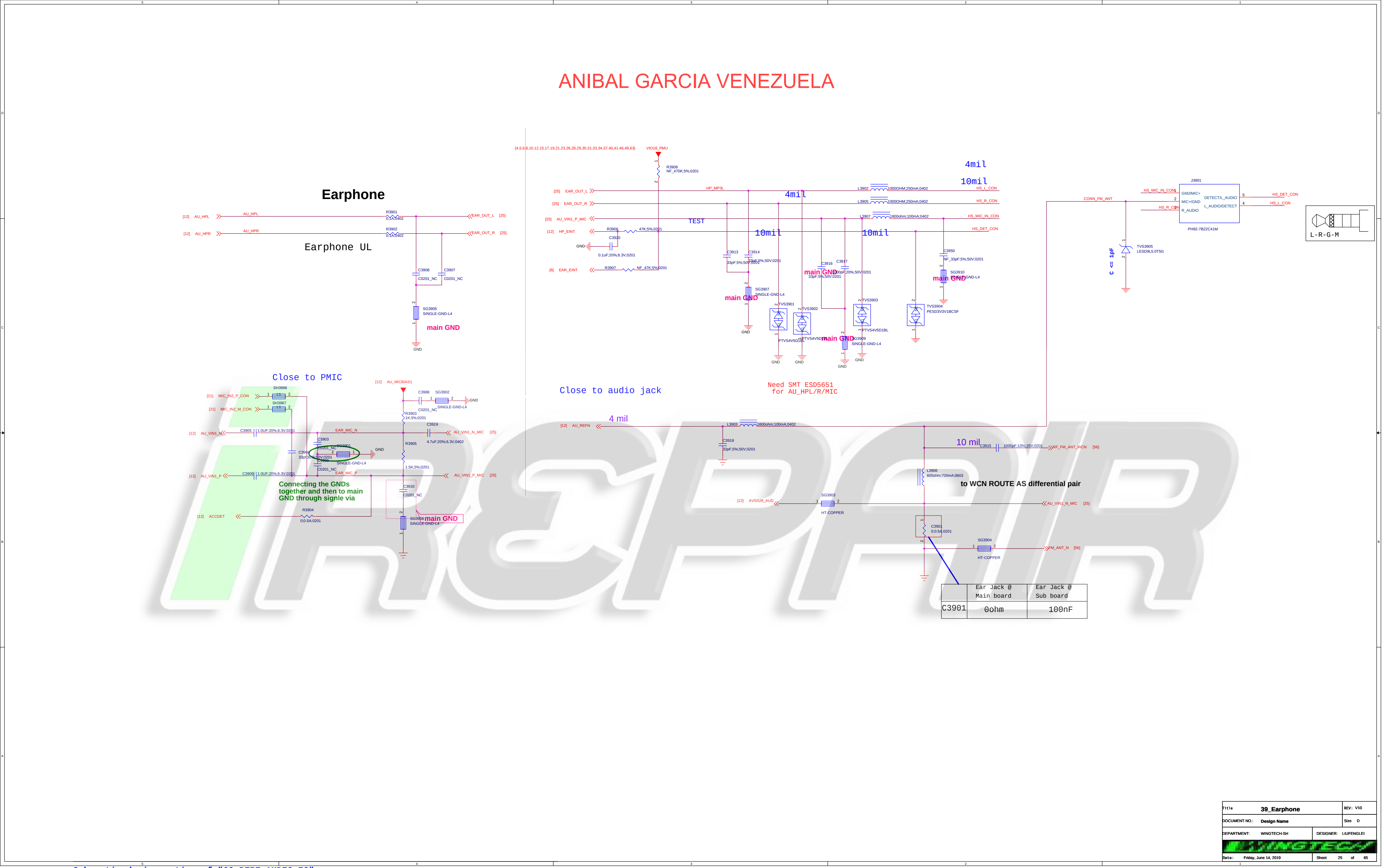


Schematic design notice of "62_PERI_AUDIO_IO" page.

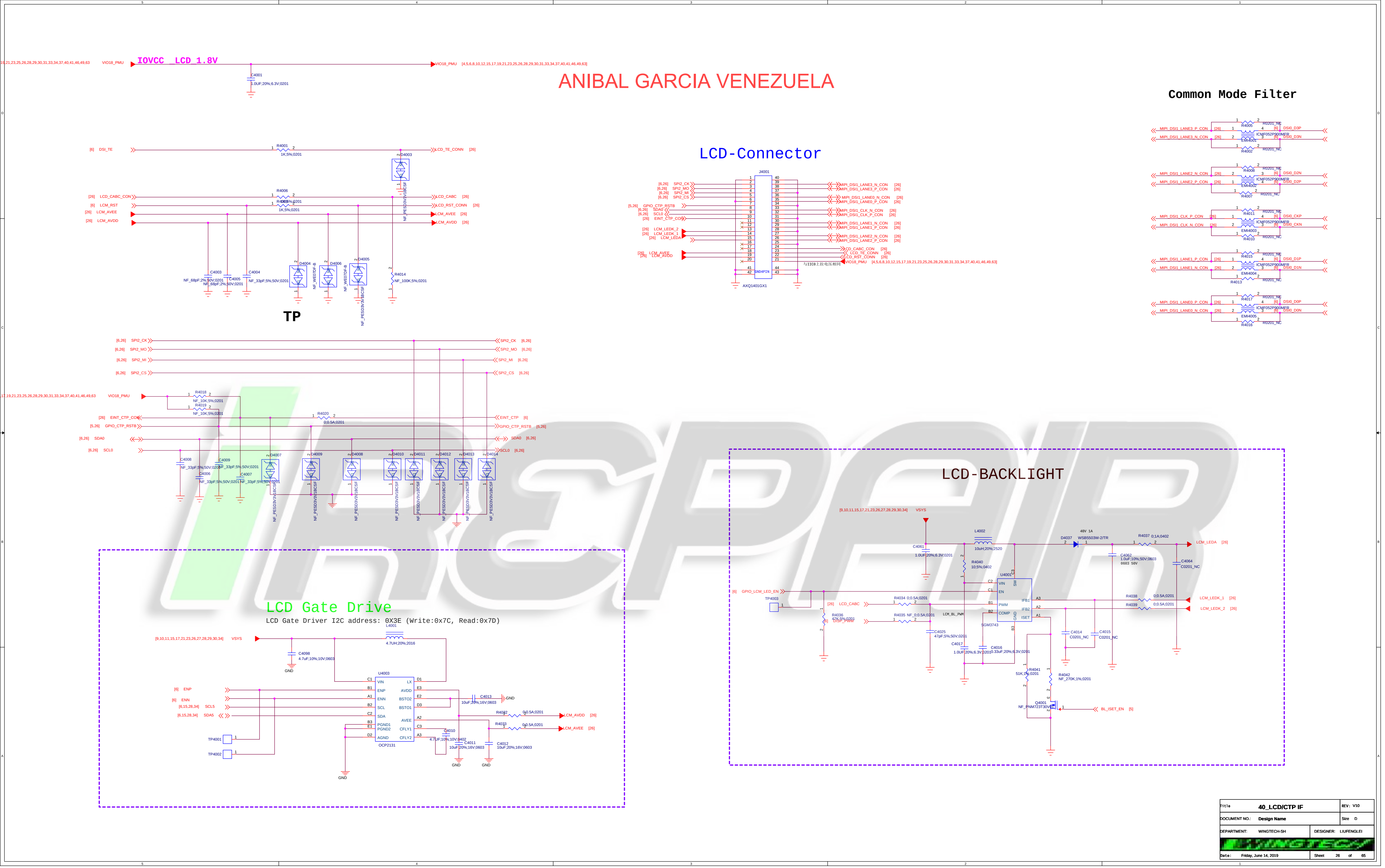
Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Note 62-3:

	Earphone Jack @ Main board	Earphone Jack @ Sub board
R6513	0 ohm	100nF

Note 62-4: Please Select ACC Mode for Operator Project to Pass Electrical MOS Test:



IOVCC LCD_1.8V
ANIBAL GARCIA VENEZUELA
Common Mode Filter
LCD-Connector
TP
LCD-BACKLIGHT
LCD Gate Drive
LCD Gate Driver I2C address: 0X3E (Write:0x7C, Read:0x7D)
Title 40_LCD/CTP IF
REV: V10
DOCUMENT NO.: Design Name
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DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
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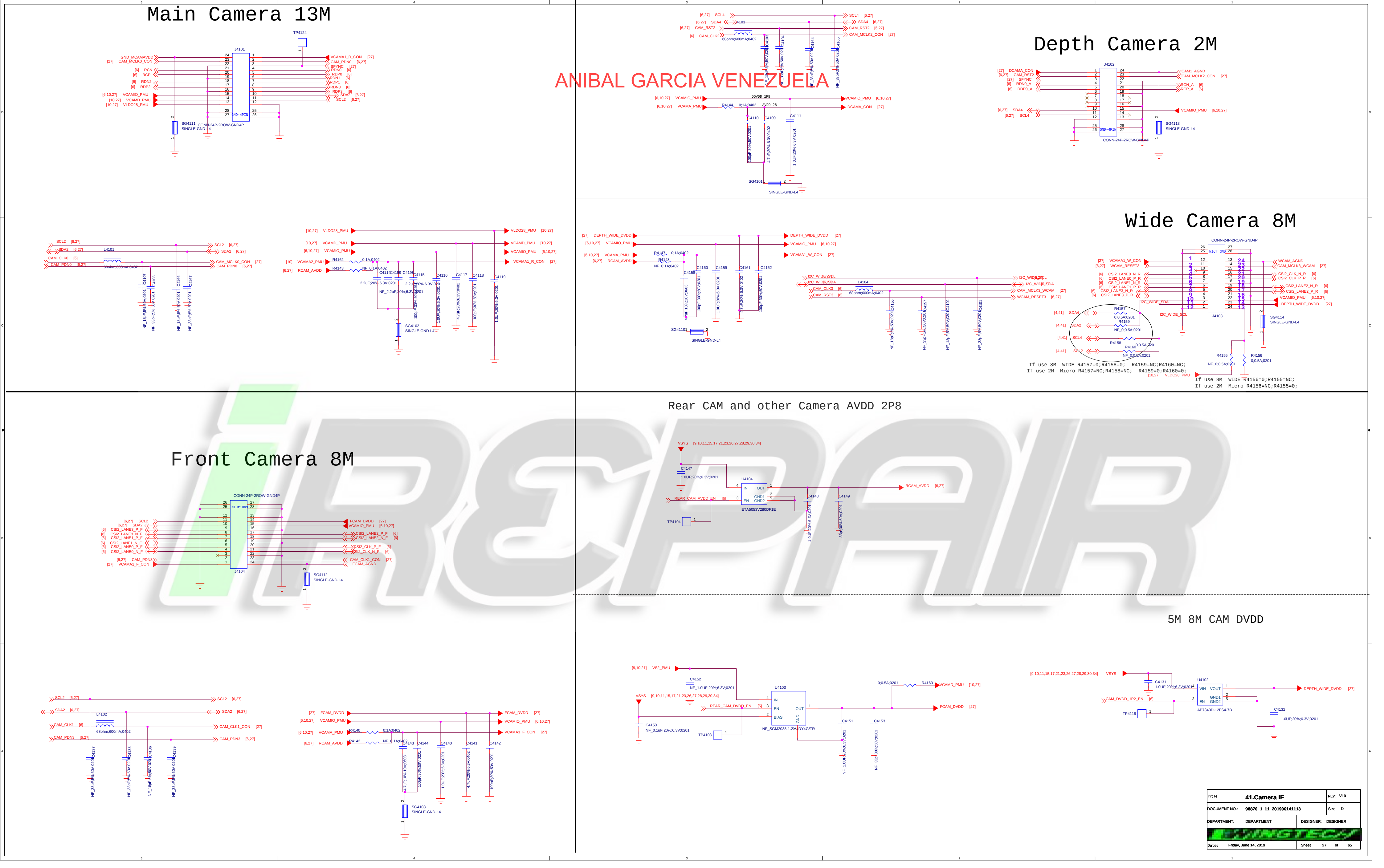

Main Camera 13M
Depth Camera 2M
ANIBAL GARCIA VENEZUELA
Wide Camera 8M
Front Camera 8M
Rear CAM and other Camera AVDD 2P8
5M 8M CAM DVDD
If use 8M WIDE R4157=0;R4158=0; R4159=NC;R4160=NC;
If use 2M Micro R4157=NC;R4158=NC; R4159=0;R4160=0;
If use 8M WIDE R4156=0;R4155=NC;
If use 2M Micro R4156=NC;R4155=0;
Title 41.Camera IF
REV: V10
DOCUMENT NO.: 98870_1_11_20190614113
Size D
DEPARTMENT: DEPARTMENT
DESIGNER: DESIGNER
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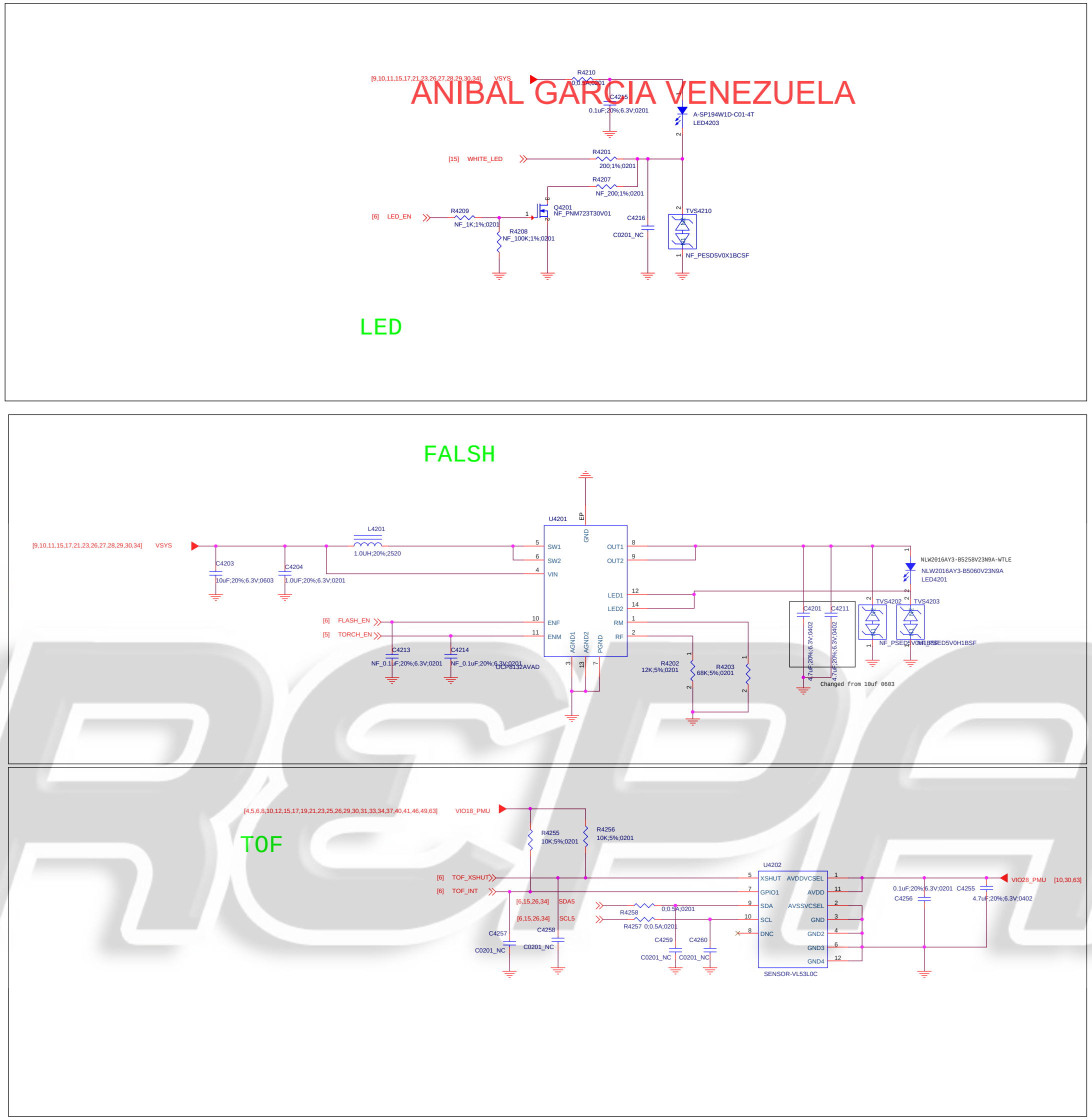

Title	42_Flash/RGB	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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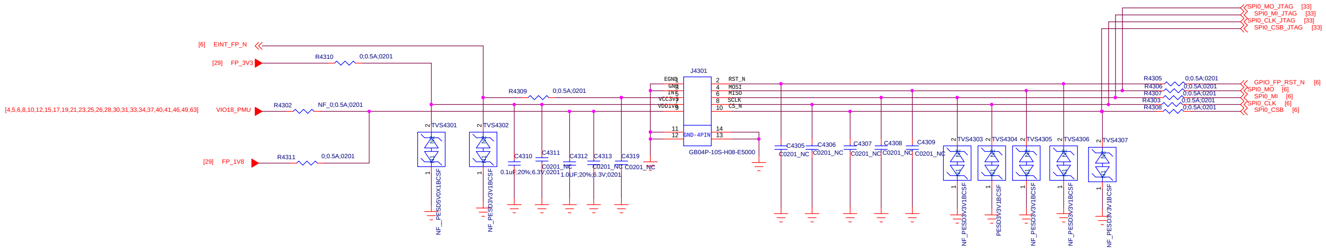
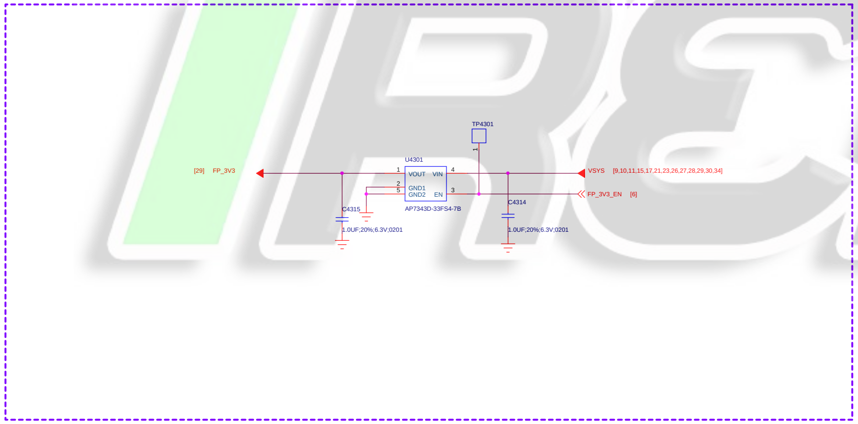
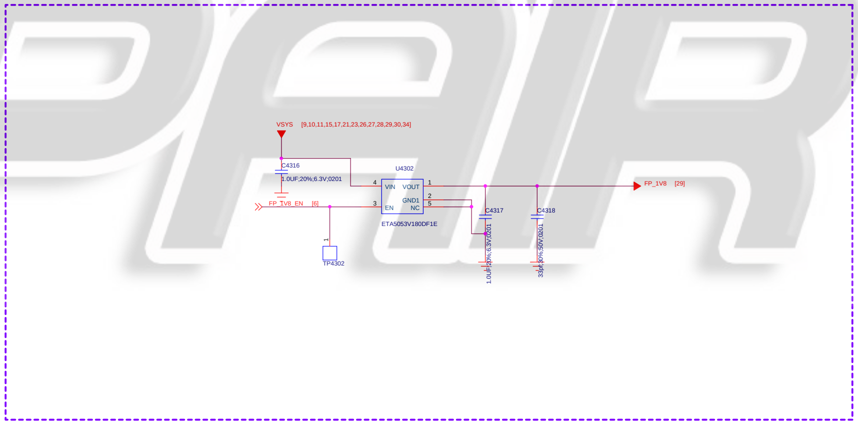
Fingerprint



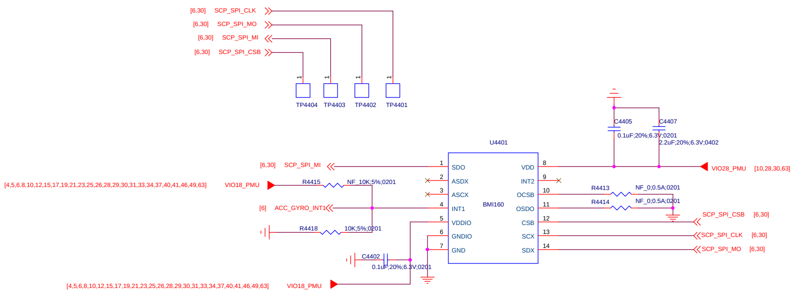
Fingerprint





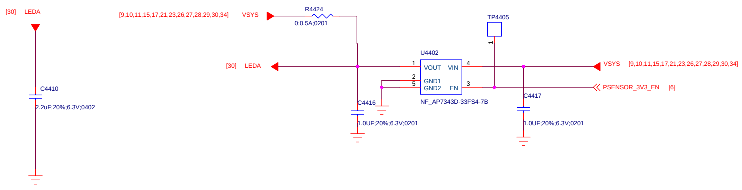
Title	43.Fingerprint	REV:	V10
DOCUMENT NO.:	98870_1_11_201906141113	Size	D
DEPARTMENT:	Wingtech-SZ	DESIGNER:	DESIGNER
Date:	Friday, June 14, 2019	Sheet	29 of 65

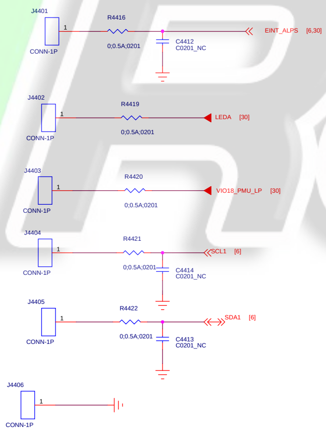
ACCELEROMETER+GYRO



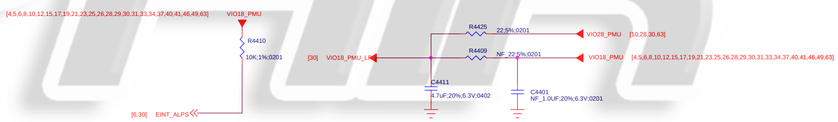
ALP-Sensor

Proximity Sensor

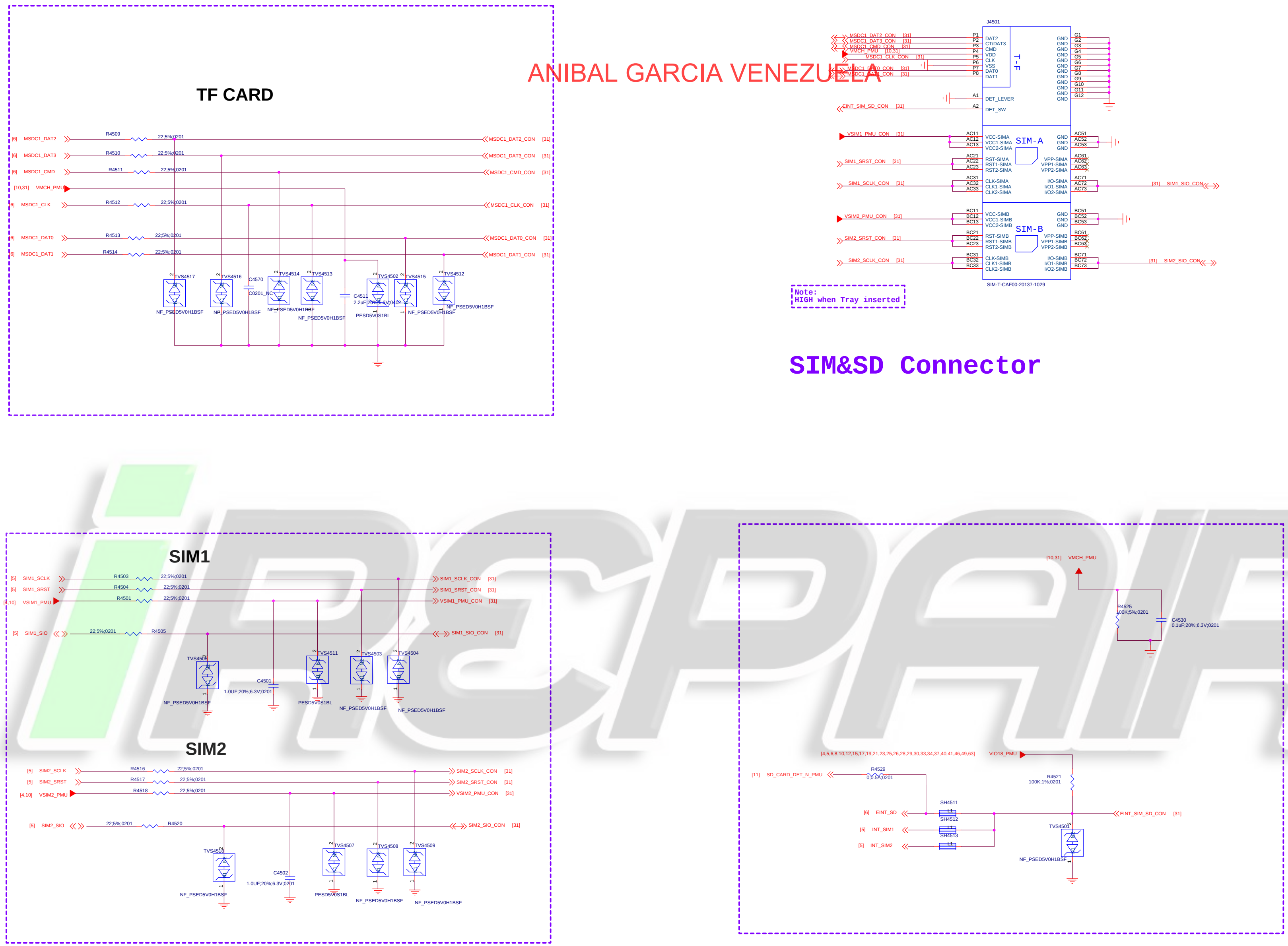




Ambient Light Sensor



Title	44_Sensor	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, June 14, 2019	Sheet 30 of 65	

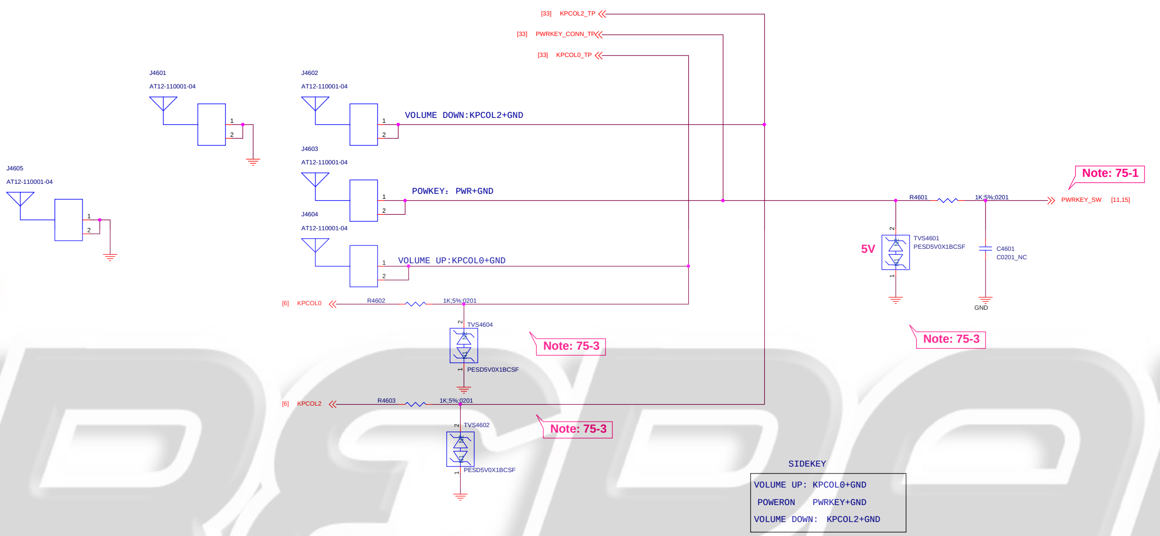


Title	45_SIM/TF IF	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, June 14, 2019	Sheet 31 of 65	

ANIBAL GARCIA VENEZUELA

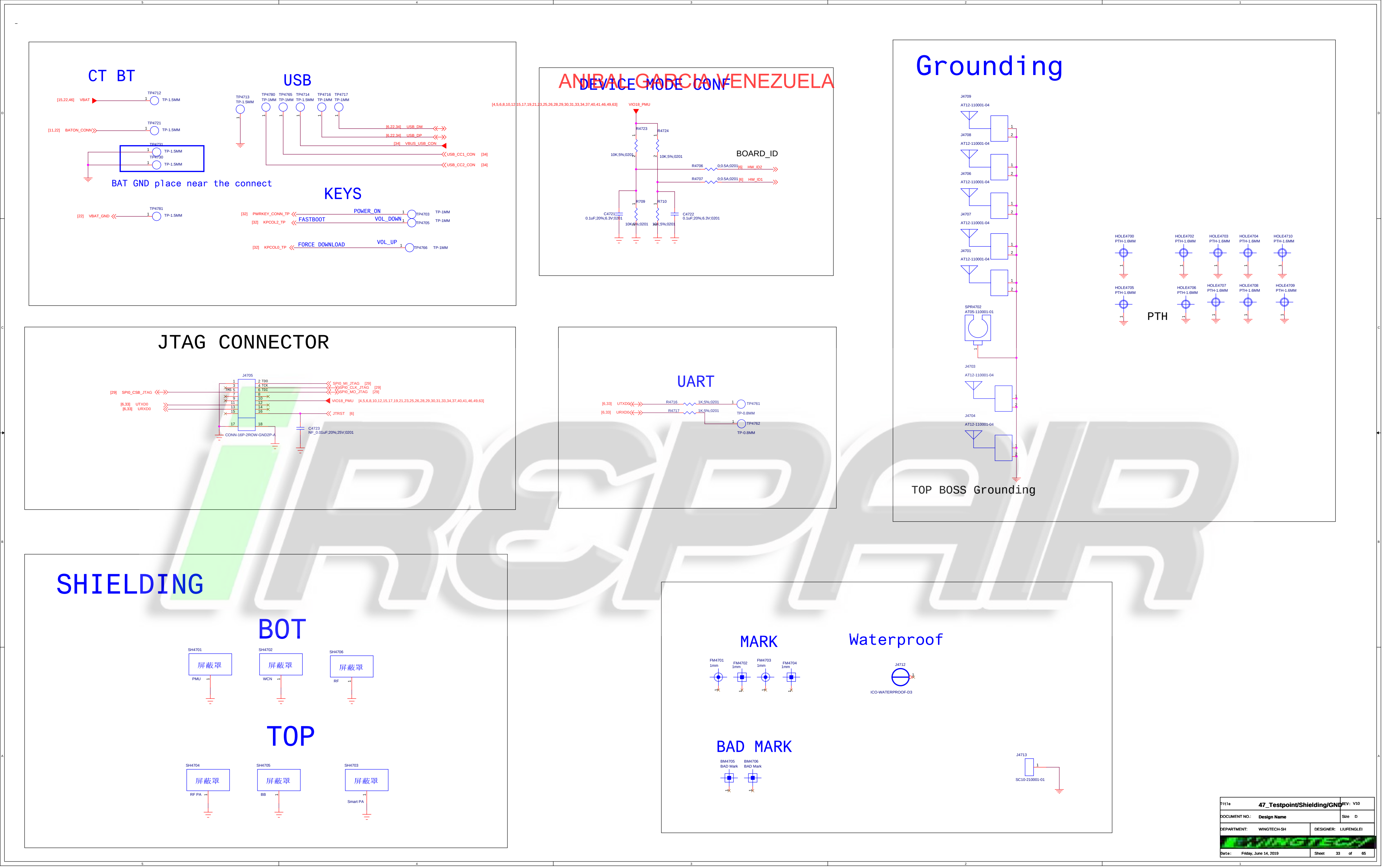
Power Key / Key Pad

DO NOT put pull-up resistor on PWRKEY



Note 75-1:DO NOT put pull-up resistor on PWRKEY
Note 75-2: Volume Up:KPCOL0+GND
Volume Down:KPCOL2+GND

Title	46_Sidekey	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, June 14, 2019	Sheet 32 of 65	


CT BT
TP4712 TP-1.5MM
[15,22,46] VBAT
TP4721 TP-1.5MM
[11,22] BATON_CONN
TP4731 TP-1.5MM
TP4730 TP-1.5MM
BAT GND place near the connect
TP4781 TP-1.5MM
[22] VBAT_GND
USB
TP4713 TP-1.5MM
TP4780 TP-1MM
TP4765 TP-1MM
TP4714 TP-1.5MM
TP4716 TP-1MM
TP4717 TP-1MM
[6,22,34] USB_DM
[6,22,34] USB_DP
[34] VBUS_USB_CON
USB_CC1_CON [34]
USB_CC2_CON [34]
KEYS
[32] PWRKEY_CONN_TP
POWER_ON
TP4703 TP-1MM
[32] KPCOL2_TP
FASTBOOT
VOL_DOWN
TP4705 TP-1MM
[32] KPCOL0_TP
FORCE DOWNLOAD
VOL_UP
TP4766 TP-1MM
DEVICE MODE CONF
VIO18_PMU
R4723
R4724
10K;5%;0201
BOARD_ID
R4706 0;0.5A;0201 [6] HW_ID2
R4707 0;0.5A;0201 [6] HW_ID1
R709
R710
C4721 0.1uF;20%;6.3V;0201
C4722 0.1uF;20%;6.3V;0201
Grounding
J4709 AT12-110001-04
J4708 AT12-110001-04
J4706 AT12-110001-04
J4707 AT12-110001-04
J4701 AT12-110001-04
SPR4702 AT05-110001-01
J4703 AT12-110001-04
J4704 AT12-110001-04
TOP BOSS Grounding
HOLE4700 PTH-1.6MM
HOLE4702 PTH-1.6MM
HOLE4703 PTH-1.6MM
HOLE4704 PTH-1.6MM
HOLE4710 PTH-1.6MM
HOLE4705 PTH-1.6MM
HOLE4706 PTH-1.6MM
HOLE4707 PTH-1.6MM
HOLE4708 PTH-1.6MM
HOLE4709 PTH-1.6MM
PTH
JTAG CONNECTOR
J4705
[29] SPI0_CSB_JTAG
[6,33] UTXD0
[6,33] URXD0
SPI0_MI_JTAG [29]
SPI0_CLK_JTAG [29]
SPI0_MO_JTAG [29]
VIO18_PMU
JTRST [6]
CONN-16P-2ROW-GND2P-A
C4723 NF_0.01uF;20%;25V;0201
UART
[6,33] UTXD0
R4716 1K;5%;0201
TP4761 TP-0.8MM
[6,33] URXD0
R4717 1K;5%;0201
TP4762 TP-0.8MM
SHIELDING
BOT
SH4701 屏蔽罩 PMU
SH4702 屏蔽罩 WCN
SH4706 屏蔽罩 RF
TOP
SH4704 屏蔽罩 RF PA
SH4705 屏蔽罩 BB
SH4703 屏蔽罩 Smart PA
MARK
FM4701 1mm
FM4702 1mm
FM4703 1mm
FM4704 1mm
Waterproof
J4712 ICO-WATERPROOF-D3
BAD MARK
BM4705 BAD Mark
BM4706 BAD Mark
J4713 SC10-210001-01
Title 47_Testpoint/Shielding/GND
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
Date: Friday, June 14, 2019
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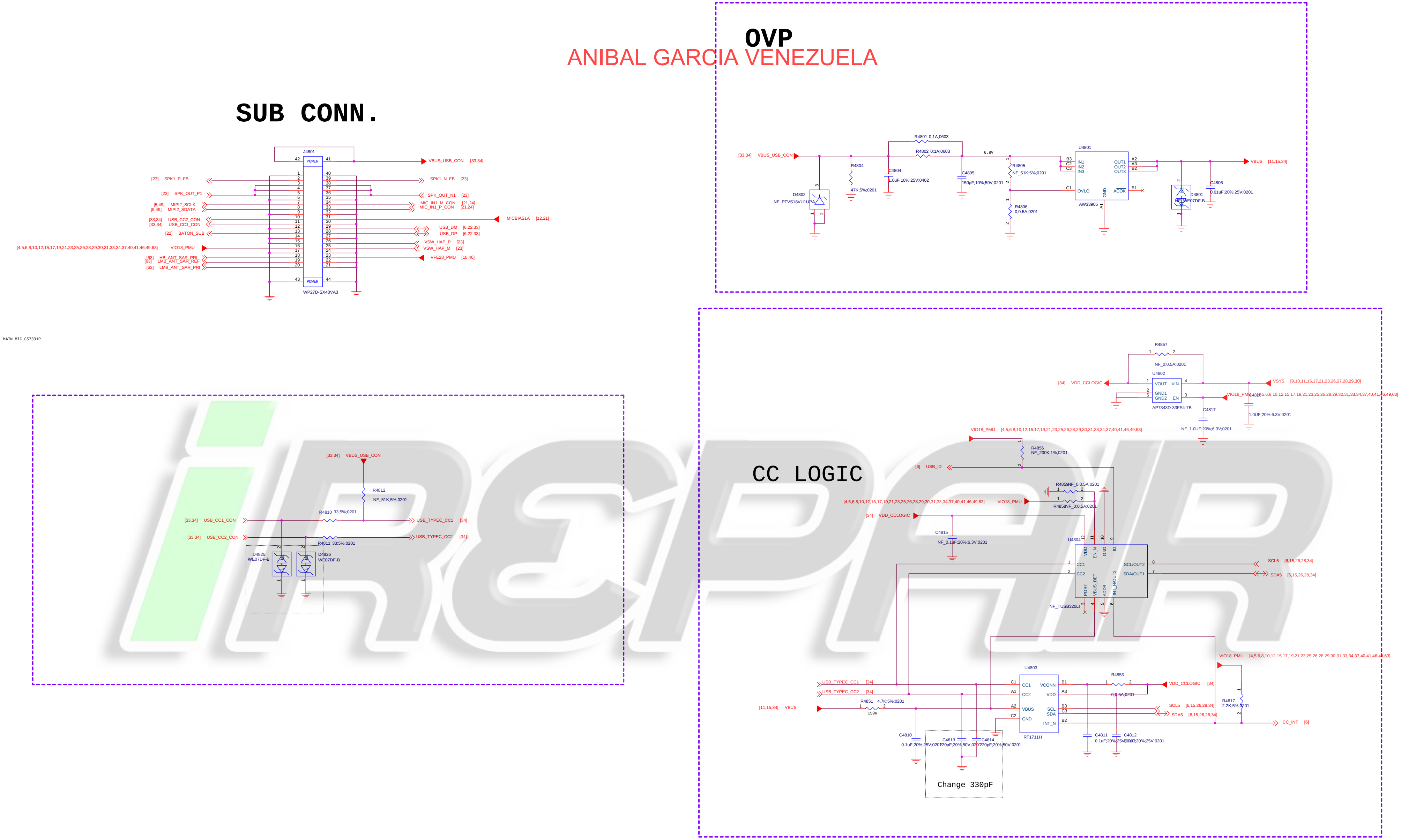

Title	48_Sub PCB IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, June 14, 2019	Sheet	34 of 65

GPIO Interface

[5,48] BPI_ANT2
[5,52] BPI_ANT1
[5,48] BPI_ANT0

[5,51] BPI_BUS7

[5,41] BPI_BUS5
[5,41] BPI_BUS4
[5,40] BPI_BUS3
[5,40] BPI_BUS2
[5,41] BPI_BUS1
[5,51] BPI_BUS0

MIPI Interface

[5,46] MIPI0_SCLK
[5,46] MIPI0_SDATA

[5,41] MIPI1_SCLK
[5,41] MIPI1_SDATA

[5,34,49] MIPI2_SCLK
[5,34,49] MIPI2_SDATA

POWER Interface

[15,22,33,46] VBAT
[4,5,6,8,10,12,15,17,19,21,23,25,26,28,29,30,31,33,34,37,40,41,46,49,63] VIO18_PMU
[10,34,46] VFE28_PMU
[10,28,30,63] VIO28_PMU
[9,46] VPA_PMU
[10,56,58] VCN28_PMU
[10,56] VCN18_PMU
[10,56] VCN33_PMU

I2C Interface

[6,23,63] SDA3
[6,23,63] SCL3

Title	Page Name = 56_RF Interface	REV:	V10
DOCUMENT NO.:	Design Name = 98870_1_11_20190614113	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Friday, June 14, 2019	Sheet	35 of 65

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Title	<Title>			
Size A1	Document Number <Doc>			Rev 1.0
Date:	Friday, June 14, 2019	Sheet	36 of 65	

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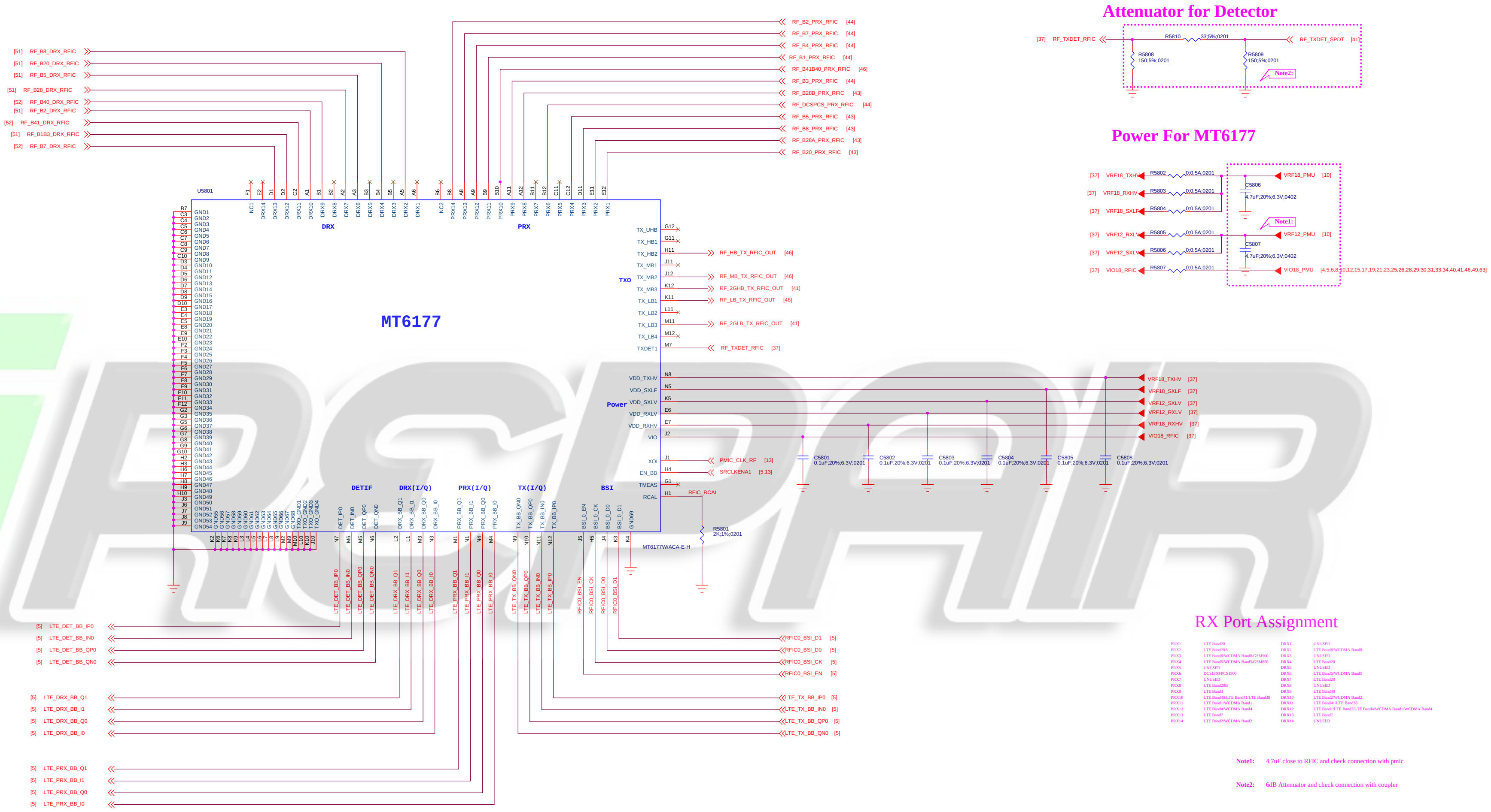

RX Port Assignment

PRX1	LTE Band20	DRX1	UNUSED
PRX2	LTE Band28A	DRX2	LTE Band8/WCDMA Band8
PRX3	LTE Band8/WCDMA Band8/GSM900	DRX3	UNUSED
PRX4	LTE Band5/WCDMA Band5/GSM850	DRX4	LTE Band20
PRX5	UNUSED	DRX5	UNUSED
PRX6	DCS1800/PCS1900	DRX6	LTE Band5/WCDMA Band5
PRX7	UNUSED	DRX7	LTE Band28
PRX8	LTE Band28B	DRX8	UNUSED
PRX9	LTE Band3	DRX9	LTE Band40
PRX10	LTE Band40/LTE Band41/LTE Band38	DRX10	LTE Band2/WCDMA Band2
PRX11	LTE Band1/WCDMA Band1	DRX11	LTE Band41/LTE Band38
PRX12	LTE Band4/WCDMA Band4	DRX12	LTE Band1/LTE Band3/LTE Band4/WCDMA Band1/WCDMA Band4
PRX13	LTE Band7	DRX13	LTE Band7
PRX14	LTE Band2/WCDMA Band2	DRX14	UNUSED

Note1: 4.7uF close to RFIC and check connection with pmic

Note2: 6dB Attenuator and check connection with coupler

Title	Page Name = 58_Transceiver	REV:	V10
DOCUMENT NO.:	Design Name = 98870_1_11_20190614113	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Friday, June 14, 2019	Sheet	37 of 65

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DOCUMENT NO.:	Design Name	Size A1
DEPARTMENT:	WINGTECH-SH	DESIGNER: LIUFENGLEI
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iREPAIR

Title	Reserved	REV: V10
DOCUMENT NO.:	Design Name	Size A1
DEPARTMENT:	WINGTECH-SH	DESIGNER: LIUFENGLEI
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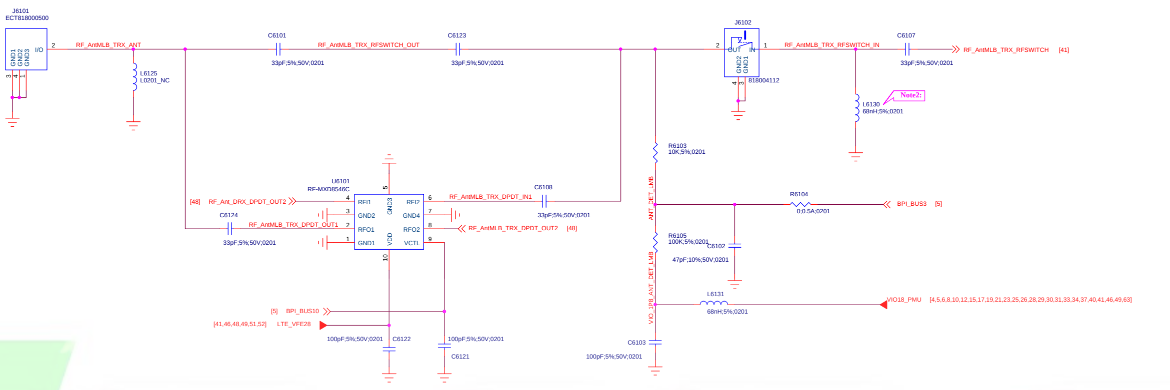
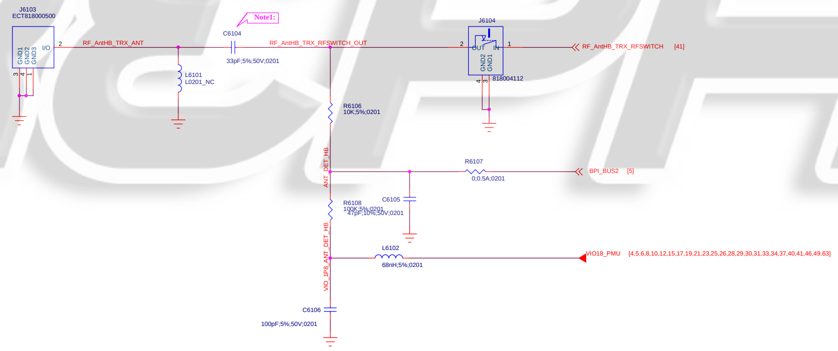
ANT_LMB



ANT_HB



Note1: RF SWITCH DETECT,MUST BE CAP IF DPDT IS WITHOUT DC BLOCK

Note2: MUST BE IND FOR RF SWITCH DETECT

Title	REV: V10
DOCUMENT NO. Design Name = 98870_1_11_201906141113	Size D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
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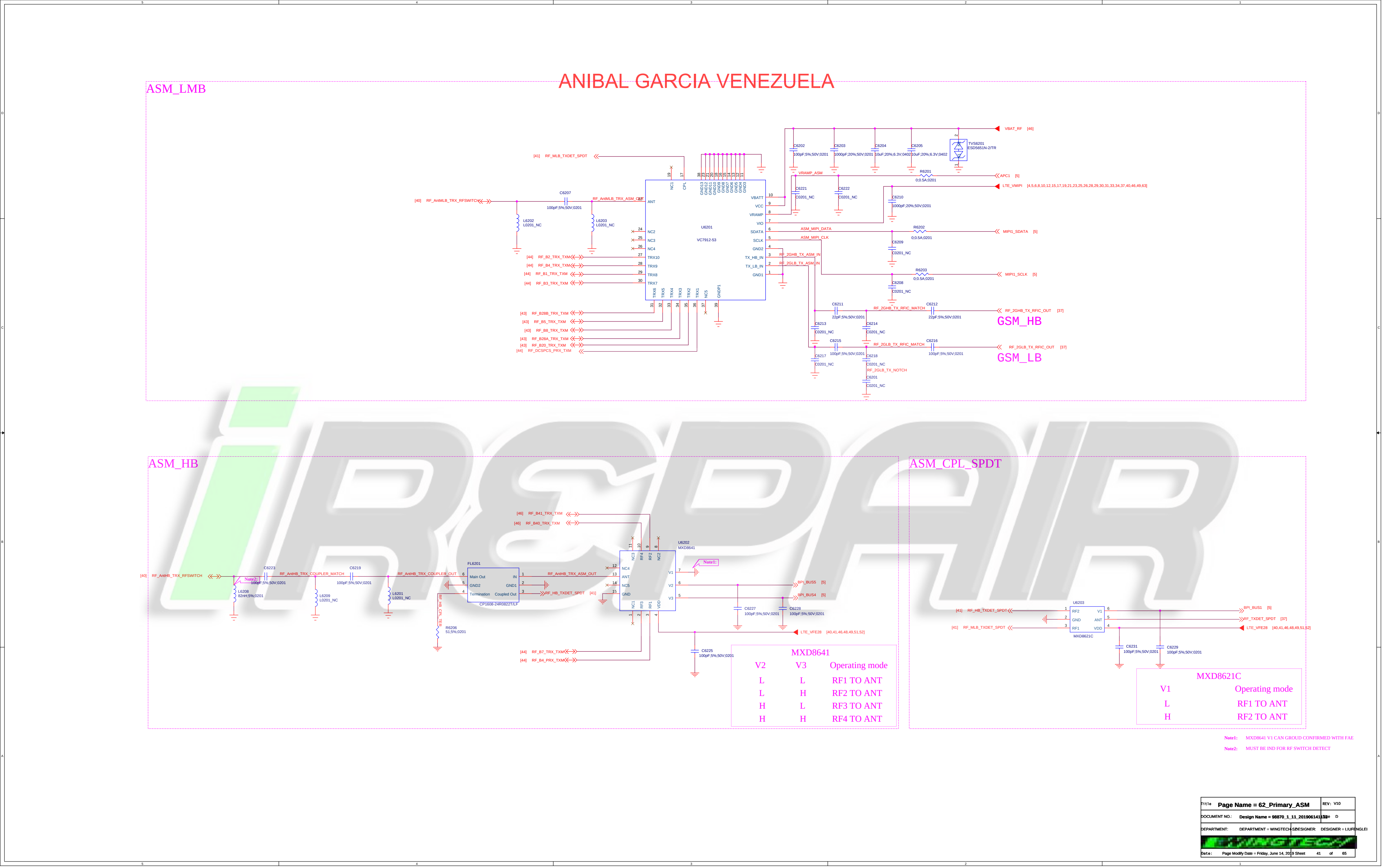

MXD8641

V2	V3	Operating mode
L	L	RF1 TO ANT
L	H	RF2 TO ANT
H	L	RF3 TO ANT
H	H	RF4 TO ANT

MXD8621C

V1	Operating mode
L	RF1 TO ANT
H	RF2 TO ANT

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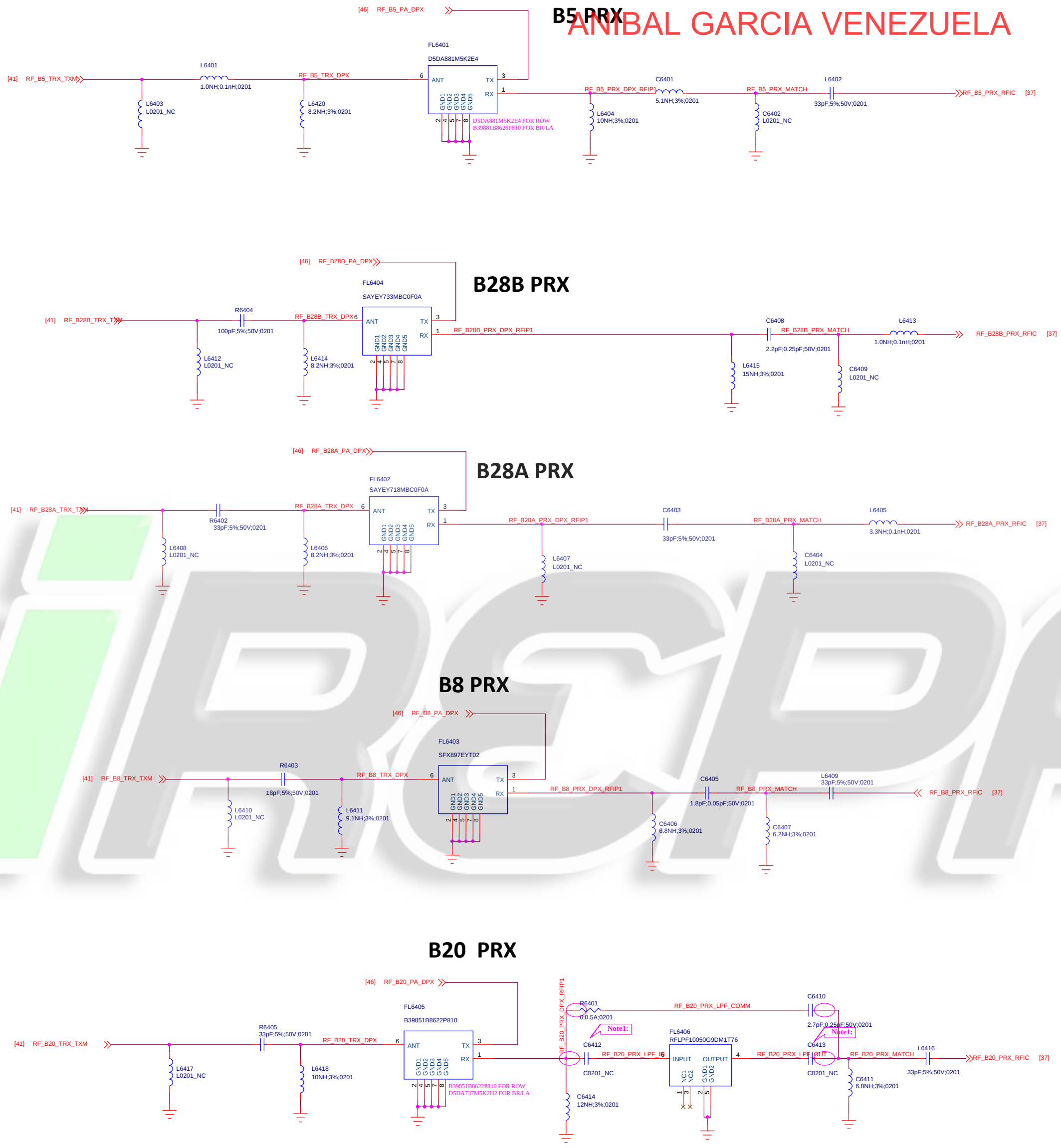
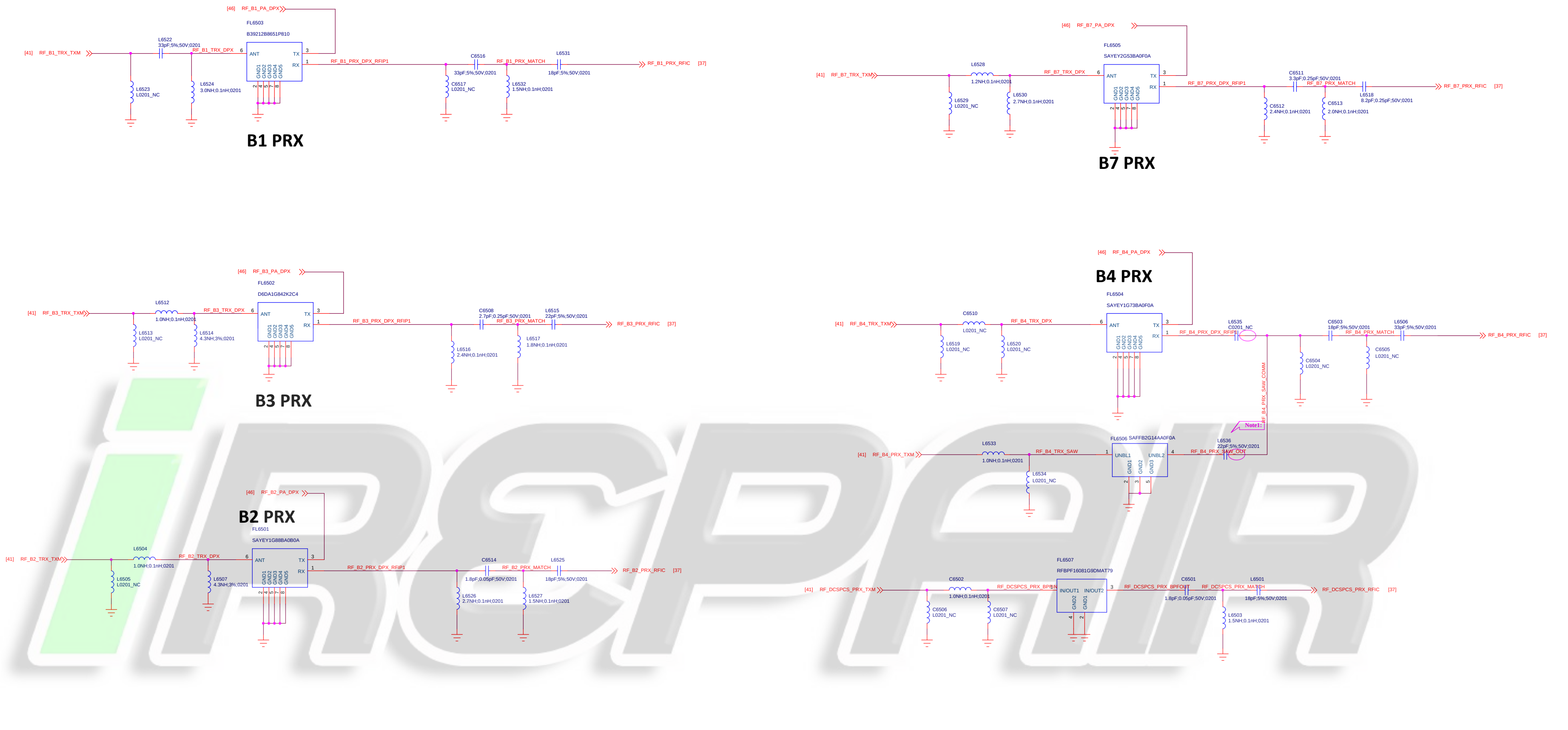
Title: <Title>		
Size: A1	Document Number: <Doc>	Rev: 1.0
Date: Friday, June 14, 2019	Sheet 42 of 65	



Note1: COMPATIBLE DESIGN FOR LPF,AND SHARE PAD IN PCB

Title	Page Name = 64_TRX_LB — REV: V10
DOCUMENT NO.:	Design Name = 98870_1_11_20190614113 — Size D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ — DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Friday, June 14, 2019 — Sheet 43 of 65

ANIBAL GARCIA VENEZUELA



Note1: COMPATIBLE DESIGN FOR B4 PRX SAW AND DUPLEXER,AND SHARE PAD IN PCB

Band Note:

	FL4704	FL4701	FL4707	FL4701	FL4702	FL4703	FL4702	FL4601	FL4602	FL4604	FL4603	U4802	U4805
China	B1	\	B3	\	G1900	\	B34/39	B5	B8	\	\	B40	B38/B41
CMCC	B1	B2	B3	\	G1900	B7	B34/39	B5	B8	\	\	B40	B38/B41
Asia Pacific Simple	B1	\	B3	\	G1900	\	\	B5	B8	\	\	B40	B38/B41
Asia Pacific	B1	\	B3	\	G1900	B7	\	B5	B8	\	\	B40	B38/B41
Overseas Full Band	B1	\	B3	B20	G1900	B7	\	B5	B8	B28A	B28B	B40	B38/B41

Title	Page Name = 65_TRX_MHB	REV:	V10
DOCUMENT NO.:	Design Name = 98870_1_11_2019061411133	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Friday, June 14, 2019	Sheet 44	of 65

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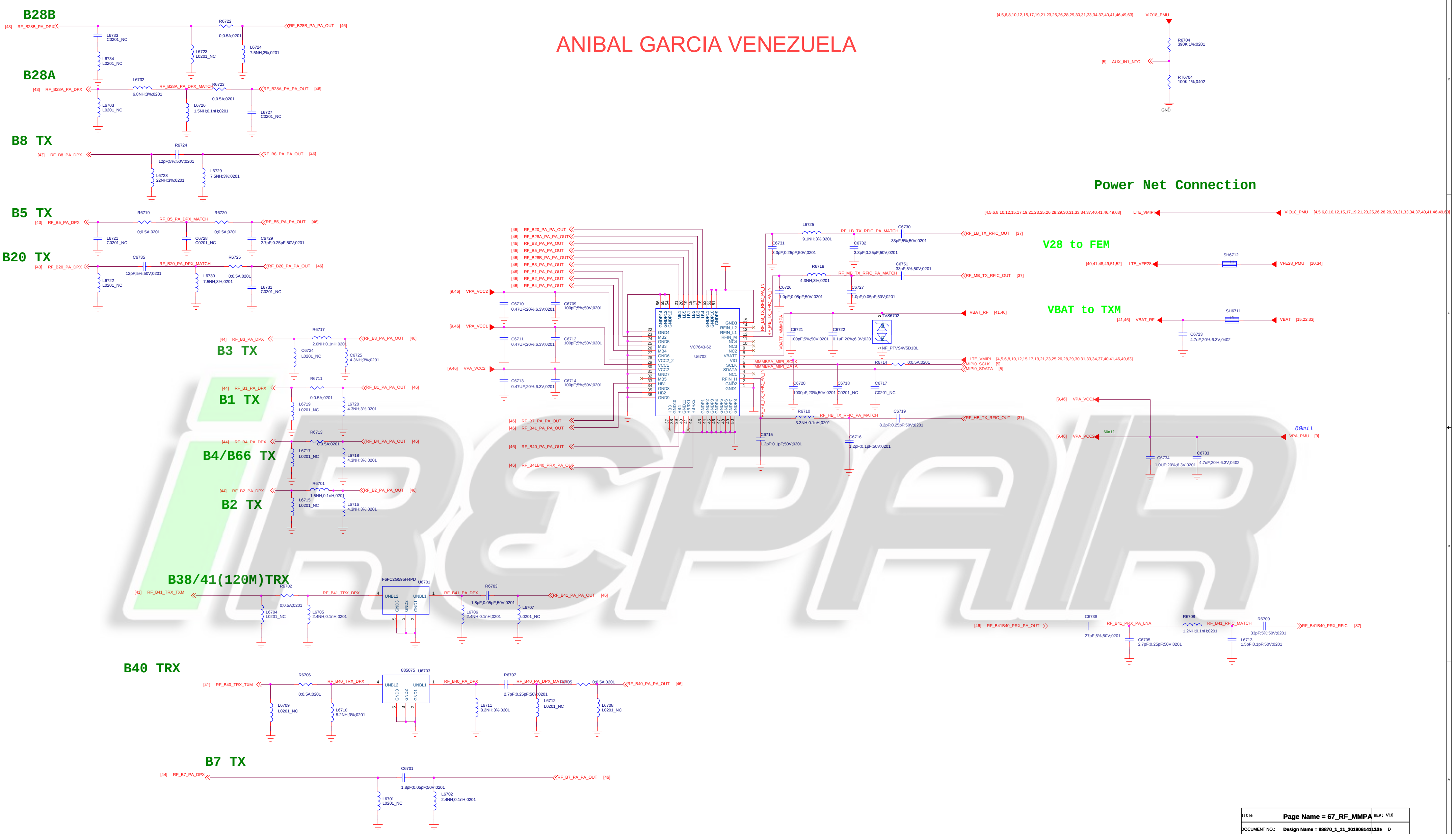
Title <Title>		
Size A1	Document Number <Doc>	Rev 1.0
Date: Friday, June 14, 2019	Sheet 45 of 65	



Title	Page Name = 67_RF_MMPA	REV:	V10
DOCUMENT NO.:	Design Name = 98870_1_11_20190614	SIZE	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
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5 4 3 2 1

D C B A

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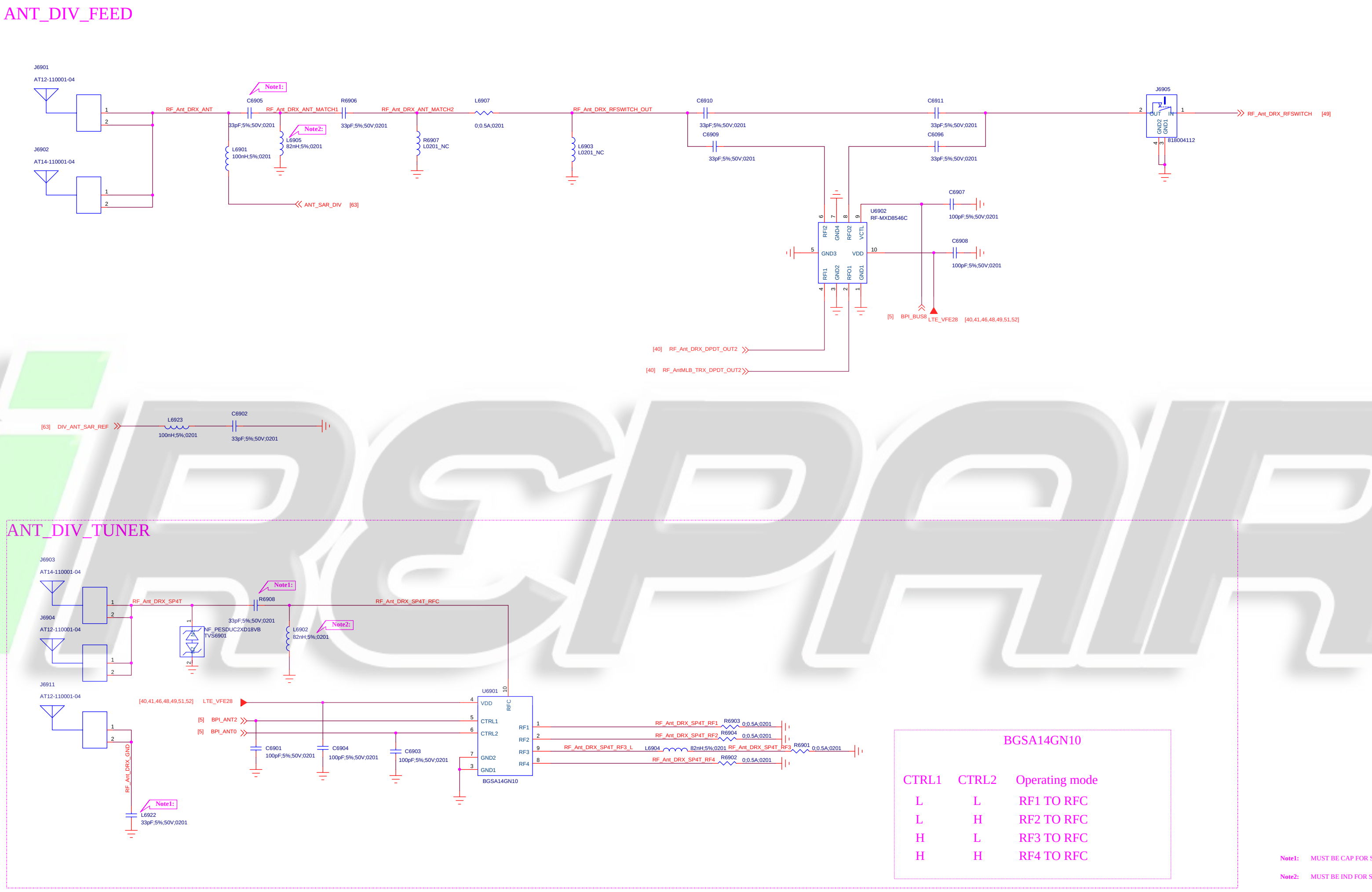
Title	<Title>	
Size A1	Document Number <Doc>	Rev 1.0
Date: Friday, June 14, 2019	Sheet 47 of 65	

5 4 3 2 1

ANIBAL GARCIA VENEZUELA



BGSA14GN10

CTRL1	CTRL2	Operating mode
L	L	RF1 TO RFC
L	H	RF2 TO RFC
H	L	RF3 TO RFC
H	H	RF4 TO RFC

Note1: MUST BE CAP FOR SAR DETECT DESIGN

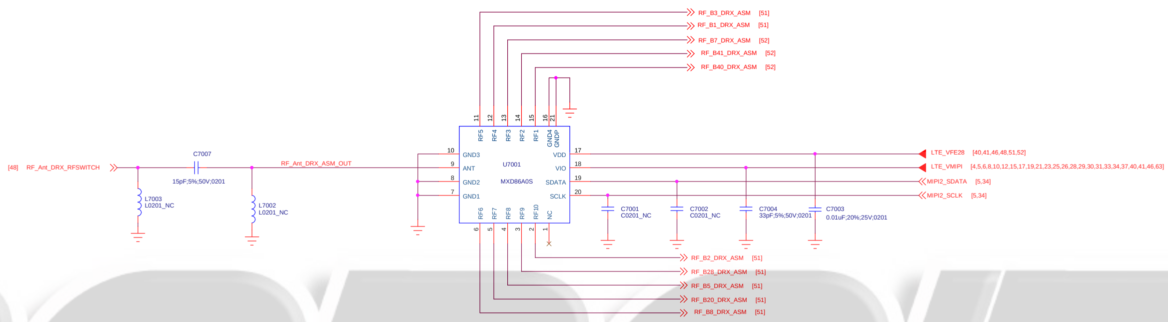
Note2: MUST BE IND FOR SAR DETECT DESIGN

Note3: 0R RES IS RESERVED FOR SAR SENSOR DESENSE,PLACE CLOSE TO ANTENNA

Note4: share pad in PCB

Title	Page Name = 69_Diversity_ANT	Rev	V10
DOCUMENT NO.:	Design Name = 98870_1_11_20190614113	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Friday, June 14, 2019	Sheet 48 of 65	

ANIBAL GARCIA VENEZUELA



Title	Page Name = 70_Diversity ASM	REV	V10
DOCUMENT NO.:	Design Name = 98870_1_11_201906141133	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
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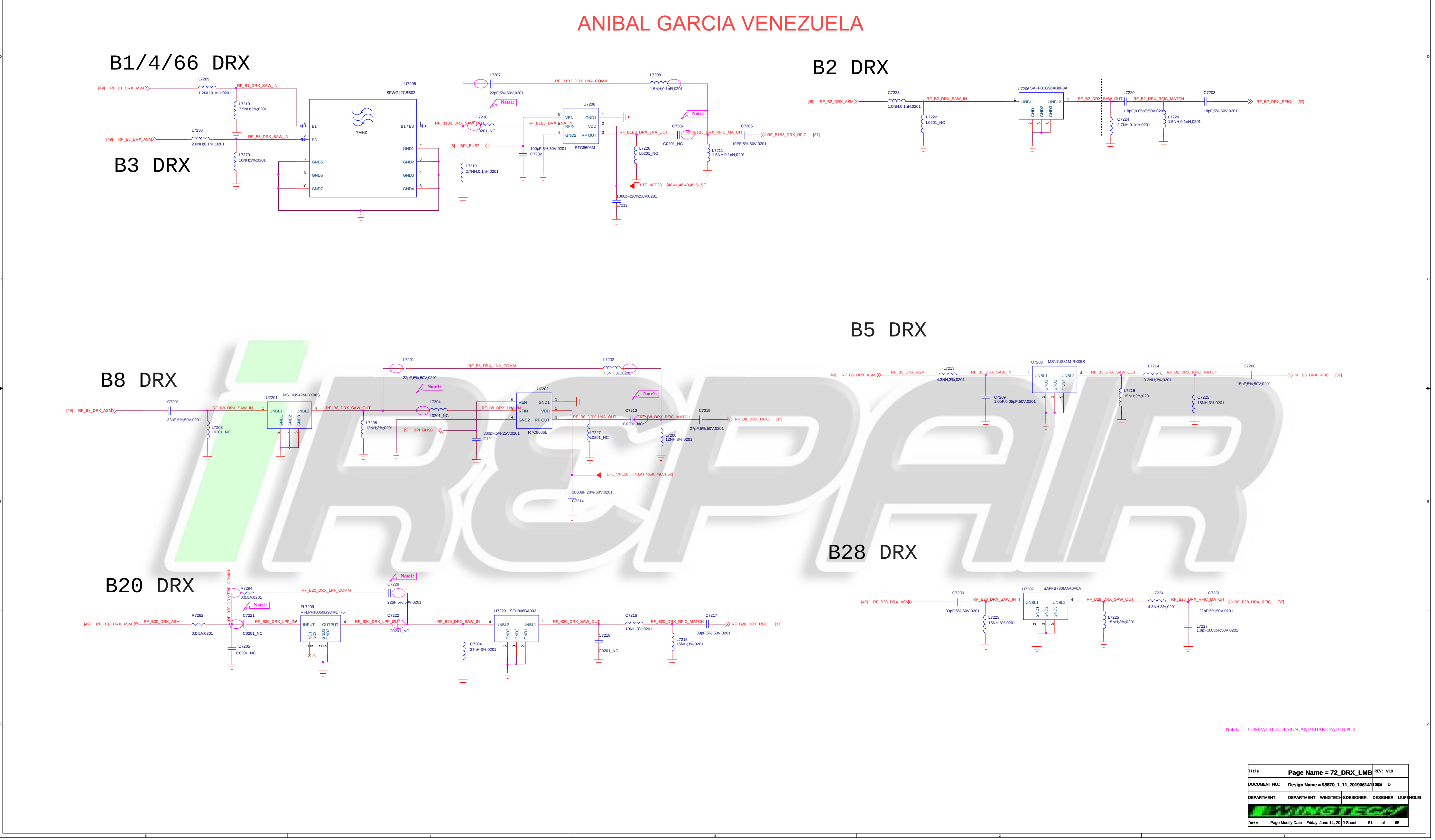
DOCUMENT NO.:	Design Name	Size A1
DEPARTMENT:	WINGTECH-SH	DESIGNER: LIUFENGLEI
Date:	Friday, June 14, 2019	Sheet 50 of 65


B1/4/66 DRX
B3 DRX
B2 DRX
B5 DRX
B8 DRX
B28 DRX
B20 DRX
Note1: COMPATIBLE DESIGN ,AND SHARE PAD IN PCB
Title
Page Name = 72_DRX_LMB
REV: V10
DOCUMENT NO.:
Design Name = 98870_1_11_20190614115
Size D
DEPARTMENT:
DEPARTMENT = WINGTECH-SZ
DESIGNER:
DESIGNER = LIUFENGLEI
Date:
Page Modify Date = Friday, June 14, 2019
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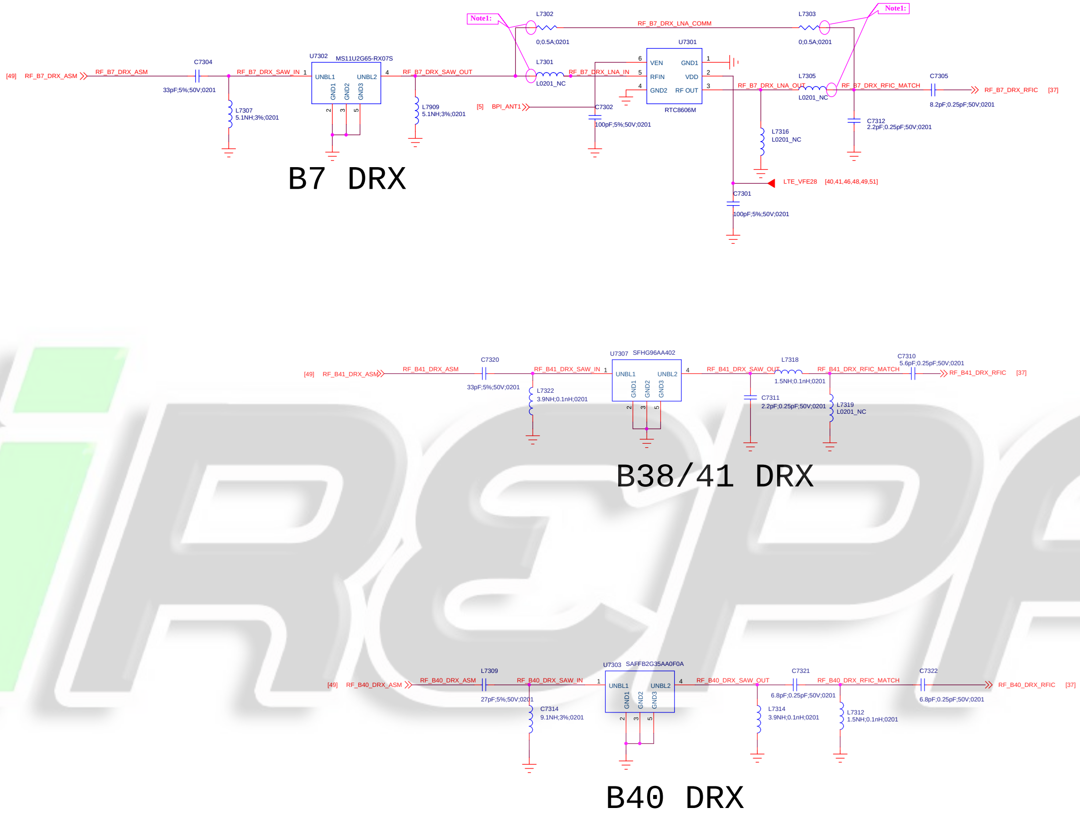

Note1: COMPATIBLE DESIGN ,AND SHARE PAD IN PCB

Title	Page Name = 73_DRX_HB	REV: V10
DOCUMENT NO.:	Design Name = 98870_1_11_20190614111	Size D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI	
Date:	Page Modify Date = Friday, June 14, 2019	Sheet 52 of 65

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Title	<Title>	
Size A1	Document Number <Doc>	Rev 1.0
Date: Friday, June 14, 2019	Sheet 53 of 65	

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iREPAIR

Title	Reserved		REV: V10
DOCUMENT NO.:	Design Name		Size A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, June 14, 2019	Sheet 54	of 65

Title	Reserved		REV: V10
DOCUMENT NO.:	Design Name		Size A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, June 14, 2019	Sheet 55 of 65	

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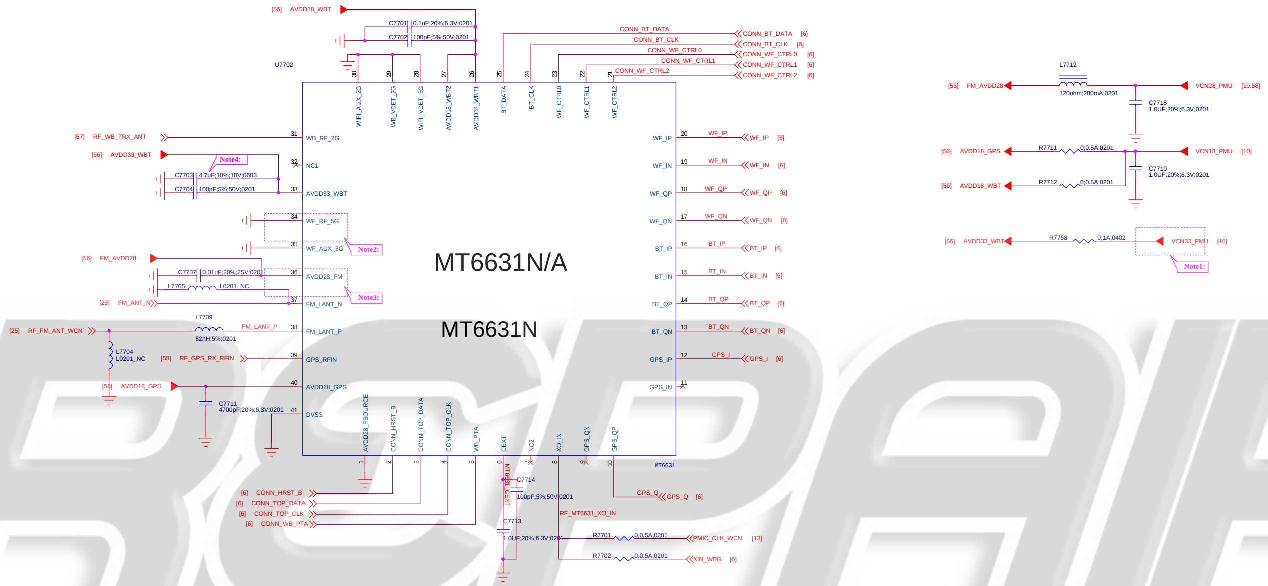

Note1: If WiFi 5G were no need, VCN33 could be chosen from PMIC output (VCN33_PMU)

Note2: If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND

Note3: Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support

Note4: 0603 package use 802167100028

Title	Page Name = 77_WCN_MT6631	REV	V10
DOCUMENT NO.:	Design Name = 98870_1_11_20190614113	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
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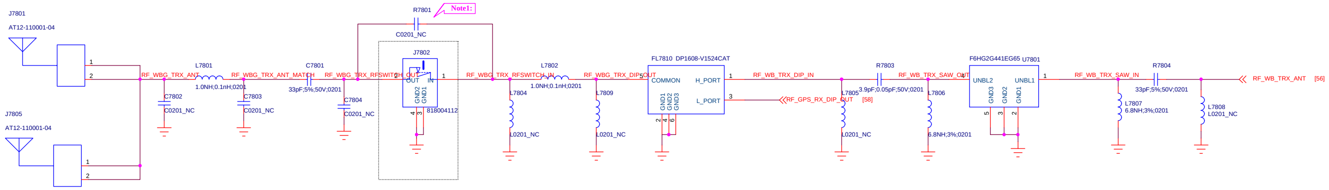
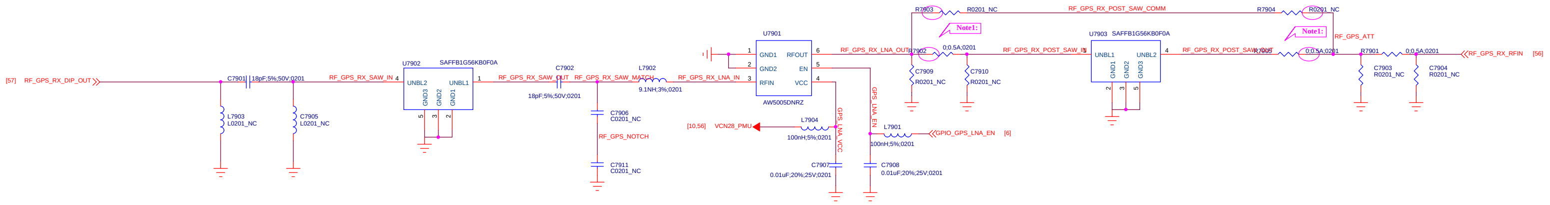
WIFI GPS ANT



Note1: COMPATIBLE DESIGN FOR WBG RF SWITCH, USE 0R AFTER MP

Title	Reserved	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, June 14, 2019	Sheet 57 of 65	



Note1: COMPATIBLE DESIGN FOR GPS POST SAW,SHARE PAD IN PCB

Title	Reserved		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size A1	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Friday, June 14, 2019	Sheet 59	of 65

Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size A1	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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Title	10_BB_ POWER_PDN	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT:	WINGTECH-SZ	DESIGNER: LIUFENGLEI
Date:	Friday, June 14, 2019	Sheet 61 of 65

If SYS CLKis used, the SYS_CLK_REQ signal needs to be connected to the host GPIO pin that configures as SRCLKENAI mode and this pin must be default low

NOTE6:

C8218/C8219 place close to XIN/XOUT Balls (less than 5mm)

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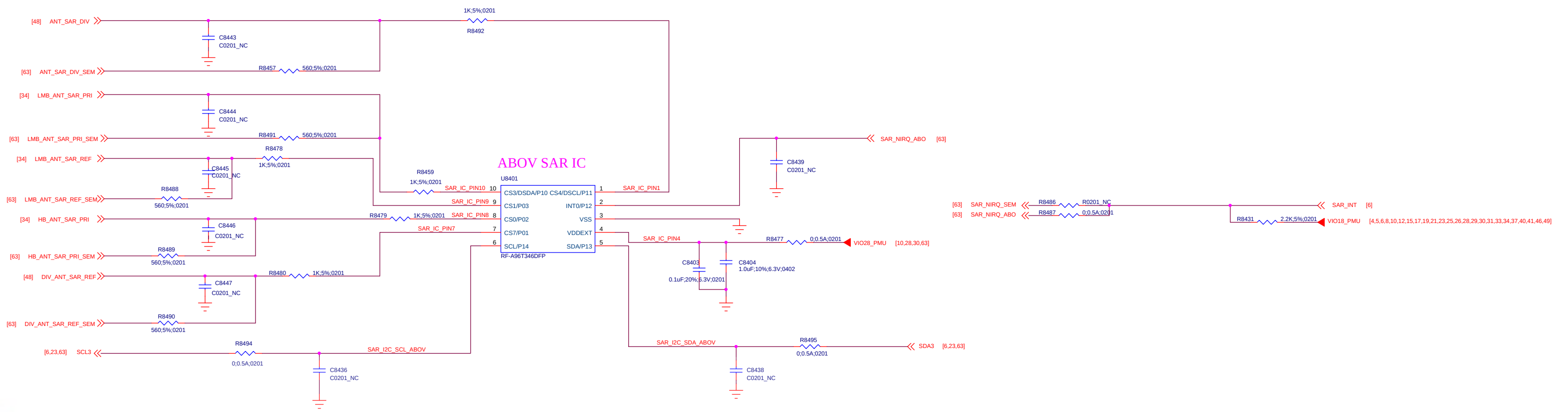
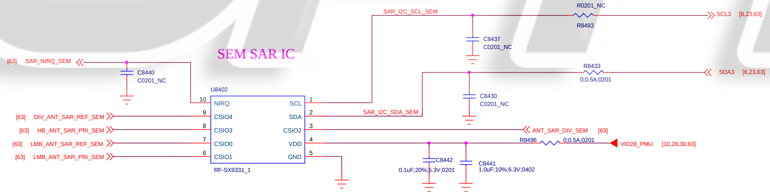
ANIBAL GARCIA VENEZUELA





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